# Interior regularity for the infinity Laplacian with bounded forcing

## 1 Statement and conventions

For a symmetric matrix $X$ and a vector $p$, write $X[p] = p^{\mathsf{T}} X p$. The unnormalized infinity Laplacian is

$$\Delta_\infty u = D^2 u[Du].$$

All balls and vector norms in this note are Euclidean. For a bounded function on a set $E$, $\operatorname{osc}_E u = \sup_E u - \inf_E u$, and

$$[g]_{\alpha,E} = \sup_{\substack{x,y\in E\\ x\neq y}} \frac{|g(x)-g(y)|}{|x-y|^\alpha}.$$

**Definition 1.1.** Let $\Omega \subset \mathbb{R}^k$ be open and $M \geq 0$. A function $u \in C(\Omega)$ belongs to $\mathcal{C}_M(\Omega)$ if, at each point of $\Omega$, every upper $C^2$ test $\phi$ satisfies $D^2\phi[D\phi] \geq -M$, and every lower $C^2$ test satisfies $D^2\phi[D\phi] \leq M$.

Here an upper test means that $u-\phi$ has a local maximum at the contact point; a constant may be added to make the two values equal. Lower tests use local minima. Tests with zero gradient are included. The notation $|\Delta_\infty u| \leq M$ below always means Definition 1.1. Our convention for $\Delta_\infty u = f$ is that upper tests give $\Delta_\infty \phi \geq f$ and lower tests give $\Delta_\infty \phi \leq f$. Thus a viscosity solution with bounded continuous right-hand side belongs to $\mathcal{C}_{\|f\|_\infty}$. No interpretation of a pointwise equation with an unspecified representative of an $L^\infty$ function is needed.

**Theorem 1.2** (Bounded forcing)**.** *For every integer $d \geq 2$ there exist $\alpha_d \in (0, 1/3]$ and $C_d < \infty$ such that, for all $R > 0$, $M \geq 0$, and bounded $u \in \mathcal{C}_M(B_R(x_0))$, one has $u \in C^{1,\alpha_d}(B_{R/2}(x_0))$ and*

$$\|Du\|_{\infty, B_{R/2}(x_0)} + R^{\alpha_d}[Du]_{\alpha_d, B_{R/2}(x_0)} \leq C_d \left\{ \frac{\operatorname{osc}_{B_R(x_0)} u}{R} + (RM)^{1/3} \right\}. \tag{1.1}$$

*The constants depend only on $d$.*

We will use the following rigidity result, whose proof is included in Sections 7–11.

**Theorem 1.3** (Liouville rigidity)**.** *Let $m \geq 1$ and $C_0 \geq 0$ be finite. Suppose that $v \in C(\mathbb{R}^m \times \mathbb{R})$ is a viscosity solution of*

$$F[v] := D^2 v[(D_x v, 1)] = 0. \tag{1.2}$$

*Suppose also that for every $(x,t) \in \mathbb{R}^m \times \mathbb{R}$ and every $S > 0$ there are $p \in \mathbb{R}^m$ and $a \in \mathbb{R}$ such that*

$$\left| v(x+h, t+\ell) - v(x,t) - p\cdot h - \left(a - \frac{|p|^2}{2}\right)\ell \right| \leq C_0 S \tag{1.3}$$

*whenever $|\ell| \leq S$ and $|h - p\ell| \leq S$. Then $v$ is affine.*

For (1.2), an upper test has $F[\phi] \geq 0$ and a lower test has $F[\phi] \leq 0$. All variables, including $t$, are tested together. The argument first reduces failure of the gradient estimate to a nonaffine solution of (1.2) satisfying (1.3). The rigidity theorem then rules it out.

## 2 Elementary viscosity facts and slope bounds

**Lemma 2.1** (Transformations and stability)**.** *The class $\mathcal{C}_M$ has the following properties.*

*(i) If $u \in \mathcal{C}_M(\Omega)$, then $-u \in \mathcal{C}_M(\Omega)$. Translations, orthogonal changes of variables, and addition of constants preserve the bound.*

*(ii) If $A, R > 0$ and $w(y) = (u(x_0 + Ry) - c)/A$, then $w \in \mathcal{C}_{R^4M/A^3}$ on its domain.*

*(iii) Locally uniform limits preserve the two inequalities when the bounds converge. More generally, the same assertion holds for continuous test-jet operators converging locally uniformly.*

*Proof.* An upper test for $-u$ becomes a lower test for $u$ after changing its sign. The value of $D^2\phi[D\phi]$ changes sign. The other invariances in (i) follow by composing test functions. For (ii), the corresponding test for $u$ has gradient $(A/R)D\phi$ and Hessian $(A/R^2)D^2\phi$, so its infinity Laplacian is $(A^3/R^4)\Delta_\infty\phi$.

For (iii), let $\phi$ be an upper test at $z$ for a limit $w$. On a sufficiently small closed ball about $z$, replace it by $\phi + \varepsilon|\cdot - z|^4$ to make the contact strict. Maxima of $w_i - \phi - \varepsilon|\cdot - z|^4$ on that ball converge to $z$ and are interior for large $i$. Apply the upper inequalities there and take the limit. The fourth-order term has zero derivatives through order two at $z$. Minima and a negative fourth-order term give the lower inequality. The same argument applies to varying continuous operators, because the evaluated jets remain in a compact set. □

For a continuous $w$ and a closed ball in its domain define

$$L_r^{\pm}(w; z) = \frac{1}{r} \max_{|h| \leq r} \{\pm(w(z + h) - w(z))\}.$$

Both quantities are nonnegative.

**Lemma 2.2** (Approximate cone comparison)**.** *If $w \in \mathcal{C}_M(\Omega)$ and $\overline{B}_R(z) \subset \Omega$, then, for $0 < r \leq R$,*

$$L_r^{\pm}(w; z) \leq L_R^{\pm}(w; z) + 3(MR)^{1/3}. \tag{2.1}$$

*Consequently $w$ is locally Lipschitz. At a fixed admissible radius the slopes are continuous in their center.*

*Proof.* First suppose $M > 0$. Set $L = L_R^+(w; z)$ and $k = (M/R^2)^{1/3}$. With $\rho = |Z - z|$, use the barrier

$$\phi(Z) = w(z) + (L + 3kR)\rho - k\rho^2.$$

At $\rho = R$ it lies above $w$, and at the center it agrees with $w$. For $0 < \rho \leq R$ its radial derivative is at least $kR$ and its second radial derivative is $-2k$. Hence

$$\Delta_\infty\phi = -2k(L + 3kR - 2k\rho)^2 \leq -2M.$$

A positive maximum of $w - \phi$ on the closed ball would occur at an interior point other than $z$. Shifting $\phi$ by a constant would then give an upper test with infinity Laplacian less than $-M$. This is impossible. Dividing $w(Z) - w(z) \leq (L + 3kR)|Z - z|$ by $r$ gives the positive case of (2.1). Apply it to $-w$ for the negative case. When $M = 0$, use the same argument with any positive upper bound for the forcing and let that bound decrease to zero.

For centers in a fixed compact neighborhood choose one common radius $R$ whose balls stay inside $\Omega$. The quantities $L_R^{\pm}$ are bounded by an oscillation divided by $R$. The preceding pointwise bound gives a common Lipschitz constant for nearby pairs. On a ball, subdivision of a line segment gives the bound for all pairs in a smaller concentric ball. Finally, continuity in $z$ follows by taking a maximum over the fixed compact set $\{|h| \le R\}$ and using uniform continuity of $w$ on a larger compact neighborhood. □

**Proposition 2.3** (Every fixed-center blow-up is linear)**.** *Let $w \in \mathcal{C}_M(\Omega)$ and $z \in \Omega$. Every sequence $r_i \downarrow 0$ has a subsequence for which*

$$V_i(Y) = \frac{w(z + r_i Y) - w(z)}{r_i}$$

*converges locally uniformly on $\mathbb{R}^k$. Every such limit is linear.*

*Proof.* Local Lipschitz continuity gives compactness on each fixed rescaled ball. The domains exhaust $\mathbb{R}^k$, so a diagonal subsequence gives a locally uniform limit. We first show that the finite limits $L_*^{\pm} = \lim_{r\downarrow 0} L_r^{\pm}(w; z)$ exist. For every fixed small $R$, (2.1) gives

$$\limsup_{r\downarrow 0} L_r^{\pm}(w; z) \le L_R^{\pm}(w; z) + 3(MR)^{1/3}.$$

Choose radii $R \downarrow 0$ that realize the lower limit. This proves equality of the upper and lower limits.

Let $h_r$ attain the positive maximum defining $L_r^{+}(w; z)$. For a fixed sufficiently small $R > r$, all the required balls are admissible once $r$ is small, and

$$L_r^{+}(w; z) \le L_r^{-}(w; z + h_r) \le L_R^{-}(w; z + h_r) + 3(MR)^{1/3}.$$

First send $r$ to zero, using fixed-radius continuity, and then send $R$ to zero. This gives $L_*^{+} \le L_*^{-}$. Changing signs proves equality; denote the common value by $L_*$.

The rescaled functions belong to $\mathcal{C}_{r_i M}$. Thus a limit $V$ is infinity-harmonic by Lemma 2.1, and

$$L_S^{+}(V; 0) = L_S^{-}(V; 0) = L_* \qquad (S > 0).$$

In particular $|V(Y)| \le L_*|Y|$. At any other center $q$,

$$L_T^{\pm}(V; q) \le \frac{L_*(T + |q|) + |V(q)|}{T}.$$

For an infinity-harmonic function, (2.1) with $M = 0$ makes the slopes nondecreasing in the radius. A fixed-radius slope at $q$ is therefore at most $L_*$ after letting $T \to \infty$. It follows that $V$ is globally $L_*$-Lipschitz.

If $L_* = 0$, then $V = 0$. Otherwise choose positive and negative extremizers $x_S, y_T$ on the balls of radii $S, T$ at the origin. Then

$$L_*(S + T) = V(x_S) - V(y_T) \le L_*|x_S - y_T| \le L_*(S + T).$$

Equality forces their norms to be $S, T$ and their directions to be opposite. Fixing first one negative extremizer and then one positive extremizer shows that all these points lie on one line: for a unit vector $e$, $V(te) = L_* t$ for all real $t$. Comparison of an arbitrary $Y$ with $te$ gives

$$L_* t - L_*|Y - te| \le V(Y) \le L_* t + L_*|Y - te|.$$

Use the lower bound as $t \to +\infty$ and the upper bound as $t \to -\infty$ to obtain $V(Y) = L_* e \cdot Y$. □

This proposition does not assert uniqueness of the blow-up slope. That conclusion will follow from the quantitative argument below.

**Lemma 2.4** (Linear lift)**.** *If $u \in \mathcal{C}_M(\Omega)$, then $U(y,b) = u(y) + b$ belongs to $\mathcal{C}_M(\Omega \times \mathbb{R})$.*

*Proof.* For an upper test $\phi$ at $(y_0, b_0)$, vertical restriction gives $\phi_b = 1$. Put $p = D_y\phi$ at the contact. If $p \neq 0$, choose an affine $\ell$ with $\ell(y_0) = b_0$ and $D\ell \cdot p = 1$. The function $\psi(y) = \phi(y, \ell(y)) - \ell(y)$ is an upper test for $u$; at the contact its gradient is $p$ and

$$D^2\psi[p] = \phi_{yy}[p] + 2\phi_{yb} \cdot p + \phi_{bb} = D^2\phi[(p,1)].$$

The required lower bound is therefore $-M$. If $p = 0$, vertical upper contact gives $\phi_{bb} \geq 0 \geq -M$, which is precisely the required bound. Lower tests give the reversed inequalities by the same restriction; when $p = 0$, use $\phi_{bb} \leq 0$. □

# 3 Three-point comparison with bounded forcing

We use the following form of the viscosity theorem of sums. For a continuous function $g$, the second-order superjet $J^{2,+}g(z)$ consists of pairs $(p, X)$ satisfying

$$g(z+h) \leq g(z) + p \cdot h + \tfrac{1}{2}X[h] + o(|h|^2).$$

Its closure $\overline{J}^{2,+}g(z)$ allows limits of jets at points tending to $z$, with convergence of the function values. Continuous test-jet inequalities extend to these closed jets: first add a positive multiple of $|h|^2$ to a jet expansion to obtain a local upper test, then let that multiple decrease to zero, and finally pass to a limit of jets.

**Lemma 3.1** (Theorem of sums)**.** *Suppose that $\sum_{j=1}^N g_j(z_j) - \Psi(z_1, \ldots, z_N)$ has a local maximum at an interior product point, with continuous $g_j$ and $C^2$ test $\Psi$. Write $A = D^2\Psi$ at that point. For every $\eta > 0$ there are symmetric $X_j$ such that*

$$(D_j\Psi, X_j) \in \overline{J}^{2,+}g_j(z_j), \qquad \operatorname{diag}(X_1, \ldots, X_N) \leq A + \eta A^2.$$

**Proposition 3.2** (Three-point estimate)**.** *Suppose $U_i \in \mathcal{C}_{M_0}(B_1 \subset \mathbb{R}^k)$, where $M_0 < \infty$ is fixed, and the $U_i$ have a common Lipschitz constant. If affine functions $\ell_i$ satisfy $\|U_i - \ell_i\|_{\infty,B_1} \to 0$, then*

$$\sup_{\substack{x,y \in B_{1/4} \\ x \neq y}} \frac{|U_i(x) + U_i(y) - 2U_i((x+y)/2)|}{|x-y|} \longrightarrow 0. \tag{3.1}$$

*Proof.* Set $w_i = U_i - \ell_i$ and $\varepsilon_i = \|w_i\|_{\infty,B_1}$. Evaluating $\ell_i$ at $\pm e/2$ for unit $e$ bounds its slope by $\operatorname{Lip}(U_i) + 2\varepsilon_i$. Thus all $w_i$ have a common Lipschitz constant $C$. If (3.1) fails, choose a subsequence and a sign such that the positive midpoint defect exceeds $\delta|x-y|$ at some pairs in $B_{1/4}$, with fixed $\delta > 0$. Changing signs preserves $\mathcal{C}_{M_0}$.

For $x, y, z \in \overline{B}_{3/4}$ with $x \neq y$, put

$$c = \frac{x+y}{2}, \quad h = z - c, \quad r = |x-y|, \qquad f(r) = r - \frac{1}{8}r^{3/2}.$$

For $0 < r \leq 3/2$ one has $\nu r \leq f(r) \leq r$, where $\nu = 1 - \sqrt{3/2}/8 > 0$. Choose a smooth $B(c) \geq \delta/2$, equal to $\delta/2$ on $\overline{B}_{1/4}$ and equal to a constant $B_{\text{out}}$ when $|c| \geq 1/2$. Maximize

$$w_i(x) + w_i(y) - 2w_i(z) - B(c)f(r) - \frac{|h|^2}{r^2}. \tag{3.2}$$

At a violating pair and $z = c$ this is positive. Everywhere it is at most

$$Cr + 2C|h| - B(c)f(r) - |h|^2/r^2 \le Cr + C^2r^2 - B(c)f(r).$$

The upper limit of this bound as $r \downarrow 0$ is at most zero, uniformly in the other variables. A positive maximum is therefore attained away from $r = 0$. Choose $B_{\text{out}} > (C + \frac{3}{2}C^2)/\nu$; the same bound excludes $|c| \ge 1/2$ at a positive maximum.

At the maximizing triple, positivity also gives $r \le 8\varepsilon_i/(\delta\nu)$, so $r \to 0$. Comparing its value with the admissible choice $z = c$ gives

$$|h|^2/r^2 \le 2(w_i(c) - w_i(z)) \le 2C|h|, \qquad |h| \le 2Cr^2.$$

Since $|c| < 1/2$, the three points are interior for large $i$.

Fix such an $i$ and its maximum. Put $q = D\ell_i$ and $e = (x - y)/r$. Apply Lemma 3.1 to $U_i(x), U_i(y), -2U_i(z)$, with

$$\Psi = \ell_i(x) + \ell_i(y) - 2\ell_i(z) + B(c)f(r) + |h|^2/r^2.$$

The derivatives of this test have the form

$$\begin{aligned} p_1 = D_x\Psi &= q + \tfrac{1}{2}f(r)DB(c) - h/r^2 + ae, \\ p_2 = D_y\Psi &= q + \tfrac{1}{2}f(r)DB(c) - h/r^2 - ae, \\ p_0 = -\tfrac{1}{2}D_z\Psi &= q - h/r^2, \\ a &= B(c)f'(r) - 2|h|^2/r^3. \end{aligned}$$

All these quantities are bounded independently of $i$, and $a \ge \delta/4$ for large $i$. The first two closed jets give $X_1[p_1] \ge -M_0$ and $X_2[p_2] \ge -M_0$. For the third jet, division by $-2$ yields the lower jet $(p_0, -X_3/2)$ of $U_i$. Hence

$$(-X_3/2)[p_0] \le M_0, \qquad X_3[p_0] \ge -2M_0.$$

Evaluating the block inequality at $\sigma = (p_1, p_2, p_0)$ gives

$$-4M_0 \le A[\sigma] + \eta|A\sigma|^2, \qquad A = D^2\Psi.$$

With $i$ and the maximum fixed, let $\eta \downarrow 0$. The matrices $X_j$ may depend on $\eta$; $A$ and $\sigma$ do not. Thus

$$-4M_0 \le A[\sigma]. \tag{3.3}$$

Compute this quadratic form along the straight path with velocity $\sigma$. At its initial point,

$$\dot r = 2a, \quad \ddot r = 0, \quad \dot c = \tfrac{1}{2}(p_1 + p_2) = O(1), \quad \dot h = p_0 - \dot c = -\tfrac{1}{2}f(r)DB(c) = O(r).$$

Since $h = O(r^2)$,

$$\frac{d^2}{d\tau^2}\frac{|h|^2}{r^2} = \frac{2|\dot h|^2}{r^2} - \frac{8\dot r\, h\cdot\dot h}{r^3} + \frac{6\dot r^{\,2}|h|^2}{r^4} = O(1).$$

Also $f''(r) = -3/(32\sqrt{r})$, and

$$\begin{aligned} \frac{d^2}{d\tau^2}(B(c)f(r)) &= B(c)f''(r)\dot r^{\,2} + 2f'(r)\dot r\, DB(c)\cdot\dot c + f(r)D^2B(c)[\dot c] \\ &= -\frac{3B(c)a^2}{8\sqrt{r}} + O(1) \longrightarrow -\infty. \end{aligned}$$

The constants hidden in $O(1)$ depend on the fixed cutoff $B$, $\delta$, and the common Lipschitz bound, but not on $i$. The affine part of $\Psi$ has zero Hessian. This contradicts (3.3). □

The order of limits in this proof is essential: the theorem-of-sums parameter tends to zero at a fixed maximum, before $i$ tends to infinity. The forcing is absorbed because its contribution to (3.3) is a fixed constant.

## 4 Uniform flatness

Fix $d \geq 2$ and put $n = d+1$. Let $\mathcal{U}$ consist of the lifts $U(y,b) = u(y) + b$, where $u \in \mathcal{C}_1(B_1^d)$, $u(0) = 0$, and $\operatorname{osc}_{B_1^d} u \leq 1$. Lemma 2.2 and the lifting lemma give uniform bounds and a common Lipschitz constant $L$ on $\overline{B}_{7/8}^n$. Put $G = B_{3/4}^n$ and $D(z) = \operatorname{dist}(z, \partial G)$. For $P \neq 0$ write $N_P = P/|P|$ and define

$$Z_\lambda(z, r; P) = \{z + srN_P + \sqrt{\lambda}\, r\xi : |s| \leq 1,\ \xi \perp P,\ |\xi| \leq 1\}.$$

A parameter $\lambda \in (0,1)$ is *good* at $(U, z, r)$ if some $P \neq 0$ satisfies

$$|U(z+h) - U(z) - P \cdot h| \leq \lambda r \quad (z + h \in Z_\lambda(z, r; P)). \tag{4.1}$$

It is bad otherwise. For $z \in G$ and $0 < r \leq D(z)/8$, let $\beta(U, z, r)$ be the supremum of the bad parameters, with value zero when there are none. Define

$$b(s) = \sup_{\substack{U \in \mathcal{U},\ z \in G \\ 0 < r \leq sD(z)}} \beta(U, z, r), \qquad 0 < s \leq 1/8.$$

The function $b$ is nondecreasing and at most one. We will use only that every $\lambda > \beta(U, z, r)$ is good; no monotonicity of goodness in $\lambda$ is asserted.

**Lemma 4.1** (Qualitative flatness)**.** *One has $b(s) \to 0$ as $s \downarrow 0$. Moreover, every fitting slope in* (4.1) *satisfies*

$$|P| \leq L + \sqrt{\lambda}, \qquad |P \cdot e_b - 1| \leq \sqrt{\lambda}. \tag{4.2}$$

*Proof.* The cylinder contains the ball of radius $\sqrt{\lambda} r$ about its center. Testing increments in every direction on this ball proves the first estimate. Testing in the lifting direction, where the increment of $U$ is exactly the increment of $b$, proves the second.

If the limit of $b$ were positive, there would be $s_i \downarrow 0$, $U_i \in \mathcal{U}$, $z_i \in G$, $0 < r_i \leq s_i D(z_i)$, and bad parameters $\lambda_i \geq \delta > 0$. Passing to a subsequence, $U_i \to U_*$ uniformly on $\overline{B}_{7/8}$ and $z_i \to z_*$ in $\overline{G}$. Let

$$a_i = r_i + |z_i - z_*| + \|U_i - U_*\|_{\infty, \overline{B}_{7/8}}, \qquad R_i = \sqrt{a_i}.$$

Then $R_i \to 0$ and each summand defining $a_i$ is $o(R_i)$. Proposition 2.3 implies, along a further subsequence, that the blow-ups of $U_*$ at $z_*$ and scales $R_i$ converge to a linear function. The same is true of

$$\widetilde{U}_i(X) = \frac{U_i(z_* + R_i X) - U_i(z_*)}{R_i}.$$

They have a common Lipschitz bound and belong to $\mathcal{C}_{R_i}$. Apply the three-point comparison to this sequence.

On the other hand, $V_i(H) = (U_i(z_i + r_i H) - U_i(z_i))/r_i$ has a subsequence converging uniformly on $\overline{B}_2$ to a function $V$. For any fixed endpoints $H_1, H_2$ in that ball their images in the $R_i$ coordinates lie in $B_{1/4}$ for large $i$. The three-point comparison, divided by $r_i/R_i$, and then with its arbitrary error tending to zero, gives

$$V(H_1) + V(H_2) = 2V((H_1 + H_2)/2).$$

Thus $V(H) = P \cdot H$. The lift gives $P \cdot e_b = 1$. Every cylinder $Z_{\lambda_i}(0, 1; P)$ is contained in $\overline{B}_{\sqrt{2}} \subset B_2$. Uniform convergence and $\lambda_i \geq \delta$ make $P$ a fitting slope for $U_i$ at $(z_i, r_i, \lambda_i)$ for large $i$, a contradiction. □

**Lemma 4.2** (Comparison of fitting slopes)**.** *Suppose two fits have error parameter $\theta^2$, where $\theta \downarrow 0$, and their slopes have norms in a fixed interval $[c_*, C_*] \subset (0, \infty)$. If they have the same center and their radii belong to $[H, 2H]$, then*

$$|P - P'| + |N_P - N_{P'}| \le C\theta, \qquad ||P| - |P'|| \le C\theta^2. \tag{4.3}$$

*The same estimates hold for equal radii $H$ if their centers differ by an axial displacement at most $H/4$ and a transverse displacement at most $\theta H/4$, measured relative to the first slope.*

*Proof.* For concentric fits, subtract the two affine approximations on the common ball of radius $\theta H$ to obtain $|P - P'| \le 3\theta$. The positive lower bound on the slope norms gives the axis estimate. The segment $z + sN_P$, $|s| \le cH$, lies in both cylinders for a sufficiently small fixed $c > 0$. Subtraction there improves the parallel estimate to $|(P - P') \cdot N_P| \le C\theta^2$. The identity

$$|P'|^2 - |P|^2 = 2P \cdot (P' - P) + |P' - P|^2$$

gives the norm estimate. For displaced centers, a ball of radius $c\theta H$ about the second center is in both cylinders. Subtract the first fit at the second center before comparing the fits on this ball. This gives the same vector estimate. A sufficiently short segment in the direction of the second axis then belongs to both cylinders, and the preceding argument applies again. □

# 5 A power estimate from rigidity

**Lemma 5.1** (Selection of scales)**.** *If no estimate $b(s) \le Bs^\gamma$ holds near zero with $B < \infty$ and $\gamma > 0$, there exist $\gamma_i \downarrow 0$, $s_i \downarrow 0$, $U_i \in \mathcal{U}$, $z_i \in G$, $0 < r_i \le s_i D(z_i)$, and bad parameters $\lambda_i$ such that*

$$\lambda_i \to 0, \qquad r_i/\lambda_i \to 0, \tag{5.1}$$

$$b(s_i)/4 < \lambda_i \le b(s_i), \qquad b(s) \le 2b(s_i)(s/s_i)^{\gamma_i} \quad (s_i \le s \le 1/8). \tag{5.2}$$

*For every fixed $K \ge 2$, for all sufficiently large $i$, every center $z$ with $|z - z_i| \le Kr_i$ and every radius $0 < r \le Kr_i$ has a good parameter $\Lambda\lambda_i$, where $\Lambda = 32$.*

*Proof.* Choose $0 < \gamma_i \le 1/2$ decreasing to zero. The function $b(s)s^{-\gamma_i}$ is unbounded near zero and bounded on each interval away from zero. On a suitable truncated interval $[h_i, 1/8]$, choose a point $s_i$ within a factor two of its supremum, so that $s_i < 1/i$ and $A_i = b(s_i)s_i^{-\gamma_i} \ge i$. Choosing the truncated supremum larger than twice the supremum on $[1/i, 1/8]$ ensures the first condition. This gives (5.2). The definition of the two suprema permits a choice of $U_i, z_i, r_i$ and an actually bad parameter with $b(s_i)/4 < \lambda_i \le b(s_i)$. By Lemma 4.1, $\lambda_i \to 0$, and

$$\frac{r_i}{\lambda_i} \le \frac{4D(z_i)s_i}{b(s_i)} \le \frac{3s_i^{1-\gamma_i}}{A_i} \longrightarrow 0.$$

For a fixed $K$, eventually $D(z) \ge D(z_i)/2$ whenever $|z - z_i| \le Kr_i$. Thus the badness at any of the stated centers and radii is at most

$$b(2Ks_i) \le 2b(s_i)(2K)^{\gamma_i} < 8\lambda_i(2K)^{\gamma_i} < 32\lambda_i < 1.$$

This proves the last assertion, including radii smaller than $r_i$. □

**Proposition 5.2** (The limiting obstruction)**.** *Failure of a power estimate for $b$ produces a nonaffine solution $v$ of* (1.2) *satisfying* (1.3) *with $C_0 = 256$ and $m = n - 1$.*

*Proof.* Take the sequence of Lemma 5.1. Choose a $\Lambda\lambda_i$ fit $P_i$ at $(z_i, r_i)$ and use orthonormal coordinates whose last axis is $P_i/|P_i|$. Put $c_i = |P_i|$. By (4.2), $1/2 \le c_i \le L+1$ for large $i$. Define $v_i$ by

$$U_i(z_i + r_i(\sqrt{\lambda_i}x, t)) = U_i(z_i) + r_i c_i\{t + \lambda_i v_i(x,t)\}. \tag{5.3}$$

The domains exhaust $\mathbb{R}^{n-1} \times \mathbb{R}$. At a test function $\phi$ for $v_i$, the corresponding gradient and Hessian for $U_i$ are

$$c_i(\sqrt{\lambda_i}D_x\phi, 1 + \lambda_i\phi_t), \qquad \frac{c_i}{r_i}\begin{pmatrix} D^2_{xx}\phi & \sqrt{\lambda_i}D^2_{xt}\phi \\ \sqrt{\lambda_i}D^2_{tx}\phi & \lambda_i\phi_{tt}\end{pmatrix}.$$

Consequently $v_i$ satisfies the two test inequalities for

$$F_i[\phi] = D^2\phi[(D_x\phi, 1 + \lambda_i\phi_t)] \tag{5.4}$$

with error at most $r_i/(\lambda_i c_i^3)$, which tends to zero.

We give the compactness details, since the shrinking transverse scale alone does not give equicontinuity. Write $\theta_i = \sqrt{\Lambda\lambda_i}$. Comparing concentric fitting cylinders of dyadically adjacent radii from $r_i$ to $Hr_i$ gives an axis change bounded by $C\theta_i(1 + \log H)$ and a norm change bounded by $C\theta_i^2(1 + \log H)$. For a fixed box $|x|, |t| \le K$, choose $H > 16K$ so large that

$$K\{\Lambda^{-1/2} + C(1 + \log H)\} < H/4.$$

For large $i$, the physical image of this box lies in the middle quarter of the fitting cylinder at $z_i$ of radius $Hr_i$. Indeed its axial displacement is at most $2Kr_i$, and its transverse displacement is at most $K\{\sqrt{\lambda_i} + C\theta_i(1+\log H)\}r_i$. Compare the radius-$Hr_i$ fit at any center in this image with the one at $z_i$, using Lemma 4.2. Next use a finite chain of concentric radii between $Hr_i$ and $Sr_i$, for any fixed $S > 0$. Lemma 5.1 supplies all these fits. If $P = (P_x, P_t)$ is the final slope, these comparisons give

$$p' = \frac{P_x}{c_i\sqrt{\lambda_i}}, \qquad q' = \frac{P_t/c_i - 1}{\lambda_i}, \qquad |p'| + |q'| \le C_{K,S}. \tag{5.5}$$

Here the estimate for $P_t - c_i$ follows from the norm estimate and $1 - N_P \cdot N_{P_i} = |N_P - N_{P_i}|^2/2$. The enclosing radius-$Hr_i$ fit also bounds $v_i$ on the fixed box.

To express the fit in these coordinates, put $A = 1 + \lambda_i q'$, $N = (A^2 + \lambda_i|p'|^2)^{1/2}$, and $\nu = (\sqrt{\lambda_i}p', A)/N$. For large $i$, $A > 0$ and $N/A = 1 + O_{K,S}(\lambda_i)$. The identity

$$(\sqrt{\lambda_i}h, \ell) = \frac{\ell N}{A}\nu + (\sqrt{\lambda_i}(h - p'\ell/A), 0)$$

shows that this increment belongs to the cylinder of radius $S$ and transverse radius $\theta_i S$ if $|\ell| \le S/2$ and $|h - p'\ell/A| \le S$, for all sufficiently large $i$. Its transverse component is at most $\sqrt{\lambda_i}S$, and its additional axial component is $O_{K,S}(\lambda_i)S$. The fit therefore implies

$$|v_i(x + h, t + \ell) - v_i(x,t) - p' \cdot h - q'\ell| \le 2\Lambda S \quad \text{if } |\ell| \le S/2,\ |h - p'\ell/A| \le S. \tag{5.6}$$

For a prescribed error, first make $S$ small and then, using the finite bound $C_{K,S}$, make $(h, \ell)$ small. This gives equicontinuity for the tail of the sequence on each fixed box. The finitely many earlier functions are individually uniformly continuous there whenever defined. The local bounds and a diagonal compactness argument give $v_i \to v$ locally uniformly, with $v(0,0) = 0$. Stability applied to (5.4) gives $F[v] = 0$.

Fix a center and a radius $R > 0$, and use (5.6) with $S = 4R$. Pass to a subsequence on which $(p', q') \to (p, q)$. Every increment with $|\ell| \le R$, $|h - p\ell| \le R$ is strictly inside the

permitted range for large $i$. Taking the limit gives (1.3) with $a = q + |p|^2/2$ and $C_0 = 8\Lambda = 256$. The coefficient subsequence may depend on the center and radius, since the function $v$ is already fixed.

Finally suppose $v(x,t) = p \cdot x + qt$. In the physical coordinates consider the slope $\widetilde{P}_i = c_i(\sqrt{\lambda_i}p, 1 + \lambda_i q)$. The cylinder $Z_{\lambda_i}(z_i, r_i; \widetilde{P}_i)$ has a bounded image under the rescaling in (5.3): its points, after division by $r_i$, are

$$s\frac{(\sqrt{\lambda_i}p, 1+\lambda_i q)}{\mu_i} + \sqrt{\lambda_i}\xi, \qquad |s|, |\xi| \le 1, \quad \mu_i = |(\sqrt{\lambda_i}p, 1+\lambda_i q)| \to 1.$$

Thus $|x| \le |p|/\mu_i + 1$ and $|t| \le 1 + \sqrt{\lambda_i}$. Uniform convergence on a fixed box gives a fitting error $o(\lambda_i r_i)$ for $\widetilde{P}_i$, contradicting badness of $\lambda_i$. Hence $v$ is nonaffine. □

Theorem 1.3 and Proposition 5.2 now imply that there exist $B < \infty$, $s_0 > 0$, and $\gamma \in (0, 1]$, depending only on $d$, such that

$$b(s) \le Bs^\gamma \qquad (0 < s \le s_0). \tag{5.7}$$

Reducing a positive exponent to at most one only weakens the estimate.

# 6 Gradient estimates and normalization

*Proof of Theorem 1.2, using Theorem 1.3.* First take $u \in \mathcal{C}_1(B_1^d)$ with $u(0) = 0$ and $\operatorname{osc} u \le 1$, and its lift $U \in \mathcal{U}$. For $z \in \overline{B}_{1/2}^n$ one has $D(z) \ge 1/4$. Choose $A_0 > B4^\gamma$ and $r_0 > 0$ so small that $4r_0 \le \min\{s_0, 1/8\}$ and $A_0 r_0^\gamma < 1$. By (5.7), $\lambda = A_0 r^\gamma$ is good for every such center and every $r \le r_0$. Put

$$r_j = 2^{-j} r_0, \qquad \rho_j = \sqrt{A_0}\, r_j^{1+\gamma/2},$$

and choose fitting slopes $P_j$ at these radii. On $\overline{B}_{\rho_j}(z)$ the error is at most $A_0 r_j^{1+\gamma}$. Subtracting the fits on the smaller common ball gives

$$|P_j - P_{j+1}| \le C r_j^{\gamma/2}.$$

Thus $P_j \to P_\infty(z)$ uniformly in $z$, with $|P_j - P_\infty| \le C r_j^{\gamma/2}$. For $\rho_{j+1} < |h| \le \rho_j$, use the $j$th fit to obtain

$$\frac{|U(z+h) - U(z) - P_\infty(z)\cdot h|}{|h|} \le C r_j^{\gamma/2} \le C'|h|^\alpha, \qquad \alpha = \frac{\gamma}{2+\gamma} \in (0, 1/3].$$

It follows that $DU(z) = P_\infty(z)$, $|DU(z)| \le L$, and the Taylor remainder estimate is uniform in $z$.

For completeness, let $z, z' \in \overline{B}_{1/2}$, $\ell = |z - z'| \le \rho_0/2$, and $|e| = 1$. Expand $U(z' + \ell e) - U(z')$ at $z'$ and also subtract its two expansions at $z$. The increments at $z$ have length at most $2\ell$. The remainder estimates give $|(DU(z') - DU(z)) \cdot e| \le C\ell^\alpha$. For larger distances use $|DU| \le L$. Hence

$$\|DU\|_{\infty, B_{1/2}} + [DU]_{\alpha, B_{1/2}} \le C_d.$$

Restriction to $(y, 0)$ gives the normalized claim for $u$.

For general $R, M, u$, set $A = \operatorname{osc}_{B_R(x_0)} u + R^{4/3}M^{1/3}$. If $A = 0$, $u$ is constant and the assertion is immediate. Otherwise

$$w(y) = \frac{u(x_0 + Ry) - u(x_0)}{A}$$

has oscillation at most one and belongs to $\mathcal{C}_{R^4M/A^3}(B_1) \subset \mathcal{C}_1(B_1)$. Rescaling its gradient estimate proves (1.1). □

We also record the normalized operator. At a nonzero gradient it is $\Delta_\infty^{\mathrm{N}}\phi = D^2\phi[D\phi]/|D\phi|^2$. The bound $|\Delta_\infty^{\mathrm{N}}u| \le M$ means the corresponding upper and lower test inequalities; at zero gradient we require $\lambda_{\max}(D^2\phi) \ge -M$ for upper tests and $\lambda_{\min}(D^2\phi) \le M$ for lower tests.

**Corollary 6.1** (Normalized operator)**.** *For the same exponent $\alpha_d$, a bounded continuous viscosity function satisfying $|\Delta_\infty^{\mathrm{N}}u| \le M$ in $B_R(x_0)$ obeys*

$$\|Du\|_{\infty,B_{R/2}(x_0)} + R^{\alpha_d}[Du]_{\alpha_d,B_{R/2}(x_0)} \le C_d\left\{\frac{\operatorname{osc}_{B_R(x_0)} u}{R} + RM\right\}. \tag{6.1}$$

*Proof.* At a center $z$, put $L = L_\rho^+(u;z)$ and $k = M + \varepsilon$. The radial barrier

$$u(z) + (L + k\rho)s - ks^2/2, \qquad s = |Z - z|,$$

has positive radial derivative for $0 < s < \rho$ and normalized infinity Laplacian $-k$. Comparing as in Lemma 2.2, then letting $\varepsilon \downarrow 0$, gives $L_r^\pm(u;z) \le L_\rho^\pm(u;z) + M\rho$. Using $\rho = R/8$ for centers in $B_{3R/4}(x_0)$ and subdividing segments gives the Lipschitz bound

$$L_0 = 8\operatorname{osc}_{B_R(x_0)} u/R + MR/8.$$

Every upper or lower test there has gradient of norm at most $L_0$, as follows by testing both directions against the Lipschitz bound. Multiplying a nonzero-gradient test inequality by $|D\phi|^2$, and using $D^2\phi[D\phi] = 0$ when $D\phi = 0$, shows that $u \in \mathcal{C}_{ML_0^2}(B_{3R/4}(x_0))$.

For every $a \in B_{R/2}(x_0)$ apply Theorem 1.2 on $B_{R/8}(a)$. Its oscillation is at most $(R/4)L_0$. With $S = \operatorname{osc} u/R + RM$ and $L_0 \le 8S$, its right-hand side is at most

$$C_d\{2L_0 + (RML_0^2/8)^{1/3}\} \le C_d' S.$$

This bounds the gradient and its local $\alpha_d$ seminorm. Pairs at distance less than $R/16$ lie in one of these smaller balls; for the remaining pairs the gradient bound supplies the same global estimate. This proves (6.1). □

*Remark* 6.2 (The upper limit on the exponent)*.* The function $u(x) = |x_1|^{4/3}$ satisfies $\Delta_\infty u = 64/81$ in the viscosity sense. Away from $x_1 = 0$ this is a direct differentiation. On $x_1 = 0$ there is no upper $C^2$ test, and every lower test has zero gradient, so its infinity Laplacian is zero and satisfies the required lower inequality. The first derivative varies as $|x_1|^{1/3}$; thus no estimate of this generality can have a Hölder exponent larger than $1/3$. The argument above does not assert that $1/3$ itself is attained.

# 7 Compactness and quadratic envelopes

We begin the proof of Theorem 1.3. Write $\mathcal{A}_{m,C_0}$ for the class of continuous functions on $\mathbb{R}^m \times \mathbb{R}$ that solve (1.2) and have the fits (1.3) at every center and radius. We may enlarge $C_0$ to $\max\{C_0, 1\}$. Constants denoted by $C$ in estimates for this class may depend on $m$ and $C_0$.

## 7.1 Changes of frame and compactness

**Lemma 7.1** (Changes of frame)**.** *Let $v \in \mathcal{A}_{m,C_0}$, $z_0 = (x_0, t_0)$, $R > 0$, and $(p_0, a_0) \in \mathbb{R}^m \times \mathbb{R}$. Then*

$$\bar v(y,s) = \frac{v(x_0 + R(y + p_0 s), t_0 + Rs) - v(x_0,t_0)}{R} - p_0 \cdot y - \left(a_0 + \frac{|p_0|^2}{2}\right) s \tag{7.1}$$

*belongs to $\mathcal{A}_{m,C_0}$. A fit $(p,a)$ for $v$ at the corresponding center and radius $R\rho$ becomes the fit $(p - p_0, a - a_0)$ for $\bar v$ at radius $\rho$.*

*Proof.* Set

$$B = \begin{pmatrix} I_m & p_0 \\ 0 & 1 \end{pmatrix}.$$

For smooth functions, at corresponding points, $\nabla_y \bar{v} = \nabla_x v - p_0$ and $D^2\bar{v} = RB^{\mathsf{T}} D^2 v B$. Since $B(\nabla_y \bar{v}, 1) = (\nabla_x v, 1)$, this gives $F[\bar{v}] = RF[v]$. Pulling back an upper or lower smooth test gives the same implication in the viscosity sense. For the fits, an increment $(h, \ell)$ in the new variables corresponds to $R(h + p_0\ell, \ell)$. The old affine increment, divided by $R$ and with the affine part of (7.1) subtracted, is

$$(p - p_0) \cdot h + \left( a - a_0 - \frac{|p - p_0|^2}{2} \right) \ell.$$

The old box conditions become $|\ell| \le \rho$ and $|h - (p - p_0)\ell| \le \rho$, and the error is $C_0\rho$. □

We say that $v$ is *centered at* $z_c$ if $v(z_c) = 0$ and $(0, 0)$ is a fit there of radius 1. Choosing $(p_0, a_0)$ to be a fit at $(z_0, R)$ in (7.1) produces a function centered at the origin. We will also center at $(0, 1)$, using a translation of the time coordinate. Put

$$L(K) = 1 + \log K \qquad (K \ge 2), \qquad \mathcal{B}_K = \{(x, t) : |x| \le K,\ |t| \le K\}.$$

Comparisons of fits across scales give the logarithmic parameter bounds below. They yield compactness after centering and uniform subquadratic spatial growth, which will confine the quadratic-envelope optimizers. The estimates use only the fits; the equation enters when the limit is identified.

**Lemma 7.2** (Bounds and compactness from fits)**.** *Suppose that $v \in \mathcal{A}_{m,C_0}$ is centered at either $(0, 0)$ or $(0, 1)$. For $K \ge 2$ one has*

$$|v(x, t)| \le CKL(K)^2 \qquad ((x, t) \in \mathcal{B}_K). \tag{7.2}$$

*Every fit $(p, a)$ at a center in $\mathcal{B}_K$, of any radius $r > 0$, satisfies*

$$|p| + |a| \le C\big(L(K) + |\log r|\big). \tag{7.3}$$

*In particular, for a center $(x, t) \in \mathcal{B}_K$,*

$$|v(x + h, t) - v(x, t)| \le Cr\big(L(K) + |\log r|\big) \qquad (|h| \le r), \tag{7.4}$$

$$|v(x, t + \tau) - v(x, t)| \le C|\tau|\big(L(K) + |\log|\tau||\big)^2 \qquad (0 < |\tau| \le \tfrac{1}{2}). \tag{7.5}$$

*For either fixed choice of center, every sequence of centered members has a locally uniformly convergent subsequence, and its limit belongs to $\mathcal{A}_{m,C_0}$ and is centered at that same point. Moreover, on every compact time interval these members have uniformly subquadratic growth in $x$:*

$$\lim_{M \to \infty} \sup_{\substack{v \text{ centered} \\ |x| \ge M,\ t \in J}} \frac{|v(x, t)|}{|x|^2} = 0 \qquad (J \Subset \mathbb{R}). \tag{7.6}$$

*Proof.* We prove the estimates for centering at the origin. Translation by one unit in time gives the other case after enlarging the boxes by a fixed factor.

First compare two fits $(p, a), (p', a')$ at the same center, with radii $R$ and $R' \in [R, 2R]$. On the horizontal slice $\ell = 0$, both estimates hold for $|h| \le R$, so $|p - p'| \le 3C_0 \le 4C_0$. Write

$$\ell_{p,a}(h, \ell) = p \cdot h + \left( a - \frac{|p|^2}{2} \right) \ell.$$

With $c = (2 + 8C_0)^{-1}$, the point $(pcR, cR)$ lies in both sheared boxes. At this point

$$(\ell_{p,a} - \ell_{p',a'})(pcR, cR) = \left(a - a' + \frac{|p - p'|^2}{2}\right) cR.$$

The two fitting errors sum to at most $3C_0R$. It follows that $|a - a'| \le 3C_0/c + 8C_0^2$. Thus the differences of both parameters are bounded by one constant whenever the radii differ by at most a factor of two. Comparing along successive dyadic radii with the centered fit of radius 1 gives, for every fit at the origin of radius $r$,

$$|p| + |a| \le C(1 + |\log r|). \tag{7.7}$$

Fix $K \ge 2$. Choose once and for all a sufficiently large constant $A = A(m, C_0)$ and set $R = AKL(K)$. If $(p_R, a_R)$ is any fit of radius $R$ at the origin, then $\mathcal{B}_K$ lies in its sheared box with

$$|t| \le R/4, \qquad |x - p_R t| \le R/4.$$

Indeed, (7.7) bounds the latter offset by $K(1 + C(1 + \log R))$, while $\log R \le \log A + 2L(K)$. Since $L(K) \ge 1$, the asserted two inequalities hold for all $K \ge 2$ once $A$ is chosen larger than a fixed multiple of $1 + \log A$. Such a choice is possible because $\log A/A \to 0$. The fit at radius $R$ now gives

$$|v(x,t)| \le C_0R + |p_R|K + \left(|a_R| + \frac{|p_R|^2}{2}\right) K \le CKL(K)^2,$$

which is (7.2).

Let $z = (x,t) \in \mathcal{B}_K$, and compare a fit $(p_z, a_z)$ of radius $R$ at $z$ with the enclosing fit $(p_R, a_R)$. For horizontal increments of length at most $R/2$ from $z$, the enclosing estimate can be subtracted at the two endpoints, with error at most $2C_0R$. Comparison with the fit at $z$ therefore gives $|p_z - p_R| \le 6C_0$. Set $c' = (4(1 + 6C_0))^{-1}$. The increments $(p_z c'R, c'R)$ from $z$ remain in both boxes: in the enclosing box the transverse offset is at most $R/4 + 6C_0c'R \le R/2$, and the time offset is at most $R/2$. The same affine-increment identity used above gives a bound on $|a_z - a_R|$ depending only on $C_0$. Hence $|p_z| + |a_z| \le CL(K)$. Dyadic comparisons at $z$, from radius $R$ to an arbitrary radius $r$, give

$$|p| + |a| \le CL(K) + C(1 + |\log(r/R)|) \le C\big(L(K) + |\log r|\big),$$

as claimed. These comparisons apply to every choice of fit.

Using a fit of radius $r$ on the horizontal slice proves (7.4). For a time increment $\tau$ with $0 < |\tau| \le 1/2$, put $B = L(K) + |\log|\tau||$ and choose the fit radius $r = A'|\tau|B$. Here $A'$ is a constant to be fixed, independent of $K$ and $\tau$. Since $B \ge 1$,

$$|\log r| \le |\log|\tau|| + \log A' + \log B \le 2B + \log A'.$$

The fit-parameter bound consequently gives $|p| \le C(3 + \log A')B$. Choose $A' \ge 1$ sufficiently large that $A' \ge C(3 + \log A')$. This choice is possible and is made before $K$ and $\tau$ vary. Then $|\tau| \le r$ and $|p\tau| \le r$, so the vertical increment lies in the fitted box. In that estimate, the affine time coefficient and error are bounded by

$$\left(|a| + \frac{|p|^2}{2}\right)|\tau| + C_0 r \le C|\tau|B^2.$$

This proves (7.5).

The two moduli, applied successively to a spatial and a time increment in a slightly larger box, give equicontinuity on compact sets. The growth bound gives local uniform boundedness. Arzelà–Ascoli and a diagonal choice of subsequence therefore give local uniform convergence. To pass an upper viscosity test to the limit, add a positive fourth-order term at its contact, preserving its first and second derivatives and making the contact strict on a small closed ball. Maxima of the sequence minus this modified test on that ball converge to the contact, so they are interior for large indices. The viscosity inequalities there pass to the limit by continuity of $F$. Subtracting the fourth-order term and using minima treats a lower test. To pass the fits, fix one center and one radius $r$. By (7.3), pass to a further subsequence of the corresponding fits with $(p_j, a_j) \to (p, a)$. For a point with $|\ell| \le r$ and $|h - p\ell| \le r$, use in the $j$th fitted box the increment $h_j = p_j\ell + (h - p\ell)$. It converges to $h$ and belongs to that closed box. Local uniform convergence gives the limiting inequality. This subsequence may depend on the fixed center and radius, which is sufficient for existence of a fit at each such pair. The centered fit passes directly to the limit. Finally, for $t$ in a fixed compact interval, (7.2) gives a bound of the form $C_J(1+|x|)(1+\log(2+|x|))^2$, uniformly in the centered class. Dividing by $|x|^2$ proves (7.6). □

If a family instead has one common prescribed value and one common fit at a fixed center and radius, one fixed change of frame (7.1) reduces it to the centered case. The corresponding local bounds and compactness hold with constants also depending on those prescribed data.

## 7.2 Quadratic envelopes and their slopes

Let $I \subseteq \mathbb{R}$ be an open interval. We say that a continuous function $f$ on $\mathbb{R}^m \times I$ has *subquadratic spatial growth on compact time intervals* if, for every $J \Subset I$,

$$\lim_{M\to\infty} \sup_{\substack{|x|\ge M \\ t\in J}} \frac{|f(x,t)|}{|x|^2} = 0.$$

For $\ell > 0$, with the indicated time argument in $I$, define

$$\begin{aligned} \mathcal{P}_\ell f(x,t) &= \sup_y \left\{ f(y, t+\ell) - \frac{|y-x|^2}{2\ell} \right\}, \\ \mathcal{M}_\ell f(x,t) &= \inf_y \left\{ f(y, t-\ell) + \frac{|y-x|^2}{2\ell} \right\}. \end{aligned} \tag{7.8}$$

Set $\mathcal{P}_0 f = \mathcal{M}_0 f = f$. The following quadratic slice comparisons will be proved directly. For a fixed center, convexity in the length will give monotone envelope slopes. The moving-center form records the cost of shifting the center; its midpoint inequality will control velocity differences in Section 9.

**Lemma 7.3** (Quadratic envelopes)**.** *Suppose that $f$ has the preceding growth property. The extrema in* (7.8) *are attained. The envelopes are continuous jointly in their center and length, including length zero wherever the time arguments remain in $I$. Whenever the expressions are defined,*

$$\mathcal{P}_\ell(\mathcal{P}_h f) = \mathcal{P}_{\ell+h} f, \qquad \mathcal{M}_\ell(\mathcal{M}_h f) = \mathcal{M}_{\ell+h} f. \tag{7.9}$$

*If $f$ is a viscosity subsolution of* (1.2)*, then, for every affine path $x(\ell) = x_0 + d\ell$ and every fixed $t$,*

$$\ell \longmapsto \mathcal{P}_\ell f(x(\ell), t) + |d|^2 \ell \log \ell \quad \textit{is convex} \tag{7.10}$$

*on its allowed range $\ell \geq 0$, with $\ell \log \ell = 0$ at zero. In particular $\ell \mapsto \mathcal{P}_\ell f(x,t)$ is convex. If $f$ is a viscosity supersolution, then $\ell \mapsto \mathcal{M}_\ell f(x,t)$ is concave. If $f_j \in C(\mathbb{R}^m \times I)$, $f_j \to f$ locally uniformly, and*

$$\lim_{M\to\infty} \sup_j \sup_{\substack{|y|\geq M \\ t\in J}} \frac{|f_j(y,t)|}{|y|^2} = 0 \qquad (J \Subset I),$$

*then $\mathcal{P}_\ell f_j \to \mathcal{P}_\ell f$ and $\mathcal{M}_\ell f_j \to \mathcal{M}_\ell f$ locally uniformly in the allowed centers and positive lengths.*

*Proof.* Fix a compact range of centers and of allowed positive lengths. Uniform subquadratic growth and the quadratic cost force all maximizers and minimizers into a common bounded spatial set. They exist there, and the maximum and minimum of a continuous function over this fixed compact set depend continuously on the parameters. The same bounded-set conclusion holds when the length varies in $(0, \ell_0]$. On that set the values of $f$ are bounded. Comparison with the competitor $y = x$ then gives $|y - x|^2 \leq C\ell$ for an optimizer as $\ell \downarrow 0$. Uniform continuity of $f$ on the compact set gives convergence to $f(x,t)$ locally uniformly, as well as $y \to x$ locally uniformly. This proves continuity at zero. The composition identities follow by optimizing first over the intermediate point and using

$$\inf_y \left\{ \frac{|y-x|^2}{2\ell} + \frac{|z-y|^2}{2h} \right\} = \frac{|z-x|^2}{2(\ell+h)}.$$

For the convexity assertion, let $g(\ell) = \mathcal{P}_\ell f(x(\ell), t)$, and let a smooth function $\psi$ touch $g$ from above at $\ell_0 > 0$. Choose an optimizer $y_0$ at that length. After a constant adjustment,

$$\Psi(y,\ell) = \psi(\ell) + \frac{|y - x(\ell)|^2}{2\ell}$$

touches $(y,\ell) \mapsto f(y, t+\ell)$ from above at $(y_0, \ell_0)$. Put $q = (y_0 - x(\ell_0))/\ell_0$, which is the horizontal gradient of $\Psi$ there. Along the direction $(q, 1)$, the Hessian of the quadratic term has value $|d|^2/\ell_0$. For example, writing $r = y - x(\ell)$, its value along a direction with $r$-velocity $q - d$ is $\ell_0^{-1}|q - d - r/\ell_0|^2 = |d|^2/\ell_0$. The viscosity inequality gives $\psi''(\ell_0) + |d|^2/\ell_0 \geq 0$. Thus every upper test for $g(\ell) + |d|^2 \ell \log \ell$ has nonnegative second derivative at positive lengths. This is the one-dimensional viscosity criterion for convexity. To see it directly, a failure of convexity on $[a, b]$ lets one place a strictly concave quadratic above the chord but below the function somewhere; the positive maximum of their difference yields an upper test with negative second derivative. Continuity at zero extends the convexity to that endpoint.

Finally, the map $\widetilde{f}(x,t) = -f(x,-t)$ exchanges subsolutions and supersolutions of (1.2), as pullback reverses the sign of $F$. It also gives $\mathcal{P}_\ell \widetilde{f}(x,t) = -\mathcal{M}_\ell f(x,-t)$. The assertion for $\mathcal{M}$ follows.

For the last convergence assertion, fix a compact range of allowed centers and positive lengths. The common subquadratic bound and the local uniform bounds confine every optimizer for $f_j$ and for $f$ to one compact spatial set, by the first argument of the proof. On that set the functions inside the supremum and infimum converge uniformly. Their extrema therefore converge uniformly on the chosen parameter range. □

For a positive allowed length define the two slopes

$$\mathcal{S}_\ell^+ f(z) = \frac{\mathcal{P}_\ell f(z) - f(z)}{\ell}, \qquad \mathcal{S}_\ell^- f(z) = \frac{f(z) - \mathcal{M}_\ell f(z)}{\ell}. \tag{7.11}$$

For a smooth function, the quantity $H_{\mathrm{sm}} = f_t + |\nabla_x f|^2/2$ satisfies

$$F[f] = (\partial_t + \nabla_x f \cdot \nabla_x) H_{\mathrm{sm}}.$$

For the exact affine model $f(x,t) = p \cdot x + (a - |p|^2/2)t + c$ in (1.3), one has $H_{\mathrm{sm}} = a$, and direct optimization gives

$$\mathcal{P}_\ell f = f + a\ell, \qquad \mathcal{M}_\ell f = f - a\ell.$$

The optimizing displacements are respectively $\ell p$ and $-\ell p$. Thus both envelope slopes equal $a$, and both optimizer velocities, directed toward increasing time, equal $p$. These identifications concern the exact affine model; for a general fitted solution, $p, a$ remain the parameters of the chosen fit. For a general smooth function, a first-order expansion of the two quadratic optimizations formally gives $\mathcal{S}_\ell^{\pm} f = H_{\mathrm{sm}} + o(1)$ as $\ell \downarrow 0$. For a smooth solution the identity for $F[f]$ transports this quantity along the direction $(\nabla_x f, 1)$; its value may depend on the integral curve. The next lemma supplies the slope statements used for continuous solutions.

The common limiting slope is allowed to be $-\infty$.

**Lemma 7.4** (Envelope slopes)**.** *For a subsolution $f$ as in Lemma 7.3, $\mathcal{S}_\ell^+ f(z)$ is nondecreasing in $\ell$ and*

$$\limsup_{\ell \downarrow 0} \mathcal{S}_\ell^- f(z) \le \lim_{\ell \downarrow 0} \mathcal{S}_\ell^+ f(z). \tag{7.12}$$

*The right side may equal $-\infty$. For a supersolution, $\mathcal{S}_\ell^- f(z)$ is nondecreasing and*

$$\limsup_{\ell \downarrow 0} \mathcal{S}_\ell^+ f(z) \le \lim_{\ell \downarrow 0} \mathcal{S}_\ell^- f(z).$$

*Consequently, for a solution both slopes are nondecreasing and*

$$\lim_{\ell \downarrow 0} \mathcal{S}_\ell^+ f(z) = \lim_{\ell \downarrow 0} \mathcal{S}_\ell^- f(z) =: H_f(z) \in [-\infty, \infty). \tag{7.13}$$

*The function $H_f$ is upper semicontinuous.*

*Proof.* Convexity of $\mathcal{P}_\ell f(z)$, with value $f(z)$ at zero, gives the first monotonicity. Let $\widetilde{z}_\ell = (y_\ell, t - \ell)$ be a backward optimizing point for $\mathcal{M}_\ell f(z)$. Using $z$ as a forward competitor from $\widetilde{z}_\ell$ gives

$$\mathcal{S}_\ell^- f(z) \le \mathcal{S}_\ell^+ f(\widetilde{z}_\ell).$$

The optimizer localization at length zero in Lemma 7.3 shows $\widetilde{z}_\ell \to z$. Fix a small $h > 0$ whose forward time argument is allowed near $z$. For all sufficiently small $\ell < h$, monotonicity bounds the right side by $\mathcal{S}_h^+ f(\widetilde{z}_\ell)$. Continuity for fixed $h$ gives an upper limit at most $\mathcal{S}_h^+ f(z)$. Now let $h \downarrow 0$ to obtain (7.12). This order of limits also covers the value $-\infty$.

Apply this conclusion to $\widetilde{f}(x,t) = -f(x,-t)$ to obtain the supersolution statements. For a solution both monotone limits exist and the two inequalities identify them. Each is bounded above by the finite slope at any fixed positive allowed length, proving the range in (7.13). Locally in $z$, choose a common small range of lengths. The common limit is the infimum of the continuous functions $\mathcal{S}_\ell^+ f(z)$ over that range, so it is upper semicontinuous. □

**Lemma 7.5** (Bounds for fitted solutions)**.** *For a member $v \in \mathcal{A}_{m,C_0}$ centered at $(0,0)$ or $(0,1)$, $K \ge 2$, $z \in \mathcal{B}_K$, and $0 < \ell \le K$, one has*

$$\mathcal{S}_\ell^+ v(z) \le CL(K)^4, \qquad \mathcal{S}_\ell^- v(z) \le CL(K)^4. \tag{7.14}$$

*Every optimizing displacement for $\mathcal{P}_K v(z)$ or $\mathcal{M}_K v(z)$ has length at most $CKL(K)^2$.*

*Proof.* By Lemma 7.4, it suffices to bound the two slopes at length $K$. Suppose an optimizing displacement has length $BK$ with $B \ge 2$. Comparing its value with the competitor having no displacement, and using (7.2) in a box of size $(B+2)K$, gives

$$\frac{B^2 K}{2} \le CBK\big(L(K) + \log B\big)^2.$$

This estimate holds for the supremum and for the infimum. Consequently $B \le C(L(K) + \log B)^2 \le 2CL(K)^2 + 2C(\log B)^2$. Choose a fixed $B_0$ so large that $2C(\log B)^2 \le B/2$ for $B \ge B_0$. The last inequality then yields $B \le 4CL(K)^2$ for $B \ge B_0$, and the same type of bound holds when $B < B_0$. This proves the displacement claim. The optimizer and its time argument now lie in a box of size $CKL(K)^2$. Applying (7.2) there and observing that $L(CKL(K)^2) \le CL(K)$ bounds the absolute values of $v$ there by $CKL(K)^4$. Dropping the nonpositive cost in the expressions for the two slopes gives (7.14). □

The criterion for the final limit is proved in Lemma 11.1: if $v \in \mathcal{A}_{m,C_0}$ and there are $(b, t_0) \in \mathbb{R}^m \times \mathbb{R}$ and $B \in \mathbb{R}$ such that

$$\mathcal{S}_\ell^+ v(b, t_0) = \mathcal{S}_\ell^- v(b, t_0) = B \qquad \text{for every } \ell > 0,$$

then $v$ is affine. The remaining argument will construct such a limit while retaining an obstruction to affinity.

# 8 An extremal horizontal defect and contact

We next turn a nonaffine member of $\mathcal{A}_{m,C_0}$ into a contact between a solution and a subsolution. A finite horizontal probability configuration consists of positions $y_1, \dots, y_N \in \mathbb{R}^m$ and weights $\theta_i > 0$ with $\sum_i \theta_i = 1$. Its mean and root mean square radius are

$$x = \sum_i \theta_i y_i, \qquad r = \left( \sum_i \theta_i |y_i - x|^2 \right)^{1/2}.$$

## 8.1 The barycentric envelope

Let $I \subseteq \mathbb{R}$ be an open interval. For a continuous function $f$ on $\mathbb{R}^m \times I$ with subquadratic spatial growth on compact time intervals, define, for $s > 0$,

$$\mathcal{C}_s f(x,t) = \sup_{\substack{N,\ \theta_i > 0,\ \sum_i \theta_i = 1 \\ \sum_i \theta_i y_i = x}} \left\{ \sum_i \theta_i f(y_i, t) - \frac{\sum_i \theta_i |y_i - x|^2}{2s} \right\}. \tag{8.1}$$

The supremum is over finite configurations, including the single point $y_1 = x$. This operation averages one horizontal slice at the fixed time $t$; $s$ controls the variance penalty. As the next lemma shows, $\mathcal{C}_s f(\cdot, t) - |\cdot|^2/(2s)$ is the least concave majorant of $f(\cdot, t) - |\cdot|^2/(2s)$.

For the following estimate, suppose additionally that some $\kappa \ge 0$ satisfies

$$\sum_i \theta_i f(y_i, t) - f(x, t) \le \kappa r \tag{8.2}$$

for every $t \in I$ and every finite horizontal probability configuration, with mean $x$ and radius $r$. Since

$$\kappa r - \frac{r^2}{2s} = \frac{\kappa^2 s}{2} - \frac{(r - \kappa s)^2}{2s},$$

the definition gives

$$0 \le \mathcal{C}_s f(x,t) - f(x,t) \le \sup_{r \ge 0} \left( \kappa r - \frac{r^2}{2s} \right) = \frac{\kappa^2 s}{2}. \tag{8.3}$$

The scalar maximum occurs at $r = \kappa s$. The extremal construction below supplies a solution $v$ and a positive $\kappa$ attaining this bound. We will prove that $\mathcal{C}_t v - \kappa^2 t/2$ is a subsolution for $t > 0$; the sandwich places it below $v$, with contact where equality holds and a strictly positive unshifted gap. The additional Jensen assumption applies only to this estimate; the next lemma uses only the stated continuity and growth hypotheses.

**Lemma 8.1** (Barycentric envelope)**.** *Let $f$ have the preceding growth property on $\mathbb{R}^m \times I$. For $s > 0$ the supremum in* (8.1) *is attained by a finite configuration, and*

$$\mathcal{C}_s f(x,t) - \frac{|x|^2}{2s} = \min_{A\in\mathbb{R}^m} \max_{y\in\mathbb{R}^m} \left\{ f(y,t) - \frac{|y|^2}{2s} - A\cdot(y-x) \right\}. \tag{8.4}$$

*Both extrema on the right are attained. Allowing any probability measure of finite second moment and mean $x$ in* (8.1) *gives the same value. The function $\mathcal{C}_s f(x,t)$ is continuous in $(x,t,s)$ for $s > 0$, and*

$$x \longmapsto \mathcal{C}_s f(x,t) - \frac{|x|^2}{2s} \quad \textit{is concave.} \tag{8.5}$$

*If $f_j \in C(\mathbb{R}^m \times I)$, $f_j \to f$ locally uniformly, and the sequence has the common growth control*

$$\lim_{M\to\infty} \sup_j \sup_{\substack{|y|\ge M \\ t\in J}} \frac{|f_j(y,t)|}{|y|^2} = 0 \qquad (J \Subset I),$$

*then $\mathcal{C}_s f_j \to \mathcal{C}_s f$ locally uniformly for $s > 0$.*

*Proof.* For fixed $(t,s)$ put $g(y) = f(y,t) - |y|^2/(2s)$ and

$$\Gamma(A;x,t,s) = \sup_y \{g(y) - A\cdot(y-x)\}.$$

The expression on the left of (8.4) is the supremum of $\sum_i \theta_i g(y_i)$ over configurations of mean $x$, because $\sum_i \theta_i |y_i - x|^2 = \sum_i \theta_i |y_i|^2 - |x|^2$. It is therefore at most $\inf_A \Gamma(A;x,t,s)$.

Fix a compact range of $(x,t,s)$ with $s > 0$. For $A \neq 0$, use $y = x - A/|A|$ in the supremum to get $\Gamma(A;x,t,s) \ge |A| - C$ on this range. The value at $A = 0$ is uniformly bounded above, since the negative quadratic part of $g$ dominates the subquadratic growth of $f$. Thus every minimizing sequence whose values are at most $\Gamma(0) + 1$ lies in a fixed bounded set of slopes. For slopes in that set, the quantity $g(y) - A\cdot(y-x)$ tends uniformly to $-\infty$ as $|y| \to \infty$. Its maximizing positions therefore lie in one compact set. It follows that $\Gamma$ is continuous on the relevant set, the minimum and maxima are attained, and their value depends continuously on $(x,t,s)$.

Let $A_0$ be a minimizing slope, and let $E$ be its nonempty compact set of maximizing positions. Then $x$ belongs to the convex hull of $E$. Otherwise strict separation gives a vector $e$ and $\delta > 0$ with $e\cdot(y-x) \ge \delta$ for all $y \in E$. If $y_h$ maximizes $\Gamma(A_0 + he;x,t,s)$ for $h > 0$, the uniform compactness of maximizers and continuity show that every limit of $y_h$ as $h \downarrow 0$ belongs to $E$. For small $h$, $e\cdot(y_h - x) \ge \delta/2$, and hence

$$\Gamma(A_0 + he;x,t,s) \le \Gamma(A_0;x,t,s) - h\delta/2,$$

contradicting minimality. The finite-dimensional convex hull theorem now represents $x$ as a convex combination of finitely many points of $E$. Discard any zero weights. Their weighted values of $g$ equal $\Gamma(A_0;x,t,s)$, proving the reverse inequality in (8.4) and finite attainment in (8.1).

The affine majorant $g(y) \le \Gamma(A_0;x,t,s) + A_0\cdot(y-x)$ can also be integrated against any probability measure of mean $x$ and finite second moment. All terms are integrable by the growth hypothesis. Thus such measures cannot increase the value. The right side of (8.4) is an infimum of affine functions of $x$, so it is concave in $x$. This is (8.5).

For the convergence assertion, on each compact range of $(x,t,s)$ the same preceding bounds on slopes and maximizing positions hold uniformly for $f_j$: local uniform convergence gives common bounds near the compact range, and the common subquadratic control handles large $y$. Both extrema in (8.4) can therefore be restricted to fixed compact sets. On these sets the expressions converge uniformly, proving the asserted local uniform convergence. □

## 8.2 A common-time subsolution rule

To test the barycentric envelope, we must combine subsolution inequalities at several spatial points sharing one time variable. Lemma 3.1 is applied after separating and then synchronizing those times; the penalty Hessian vanishes on the direction with time speed 1 in every component, which is the direction used by $F$.

**Lemma 8.2** (Common-time subsolutions)**.** *Let $v$ be a continuous viscosity subsolution of (1.2) on $\mathbb{R}^m \times I$, and fix positive weights $\theta_i$ with $\sum_{i=1}^N \theta_i = 1$. Define*

$$V((y_i)_{i=1}^N, t) = \sum_{i=1}^N \theta_i v(y_i, t).$$

*If a $C^2$ function $\Phi$ touches $V$ from above at $(Y_0, t_0)$, where $Y = (y_i)_{i=1}^N$, then*

$$D^2\Phi(Y_0, t_0)\left[\left((\theta_i^{-1}\nabla_{y_i}\Phi(Y_0, t_0))_{i=1}^N, 1\right)\right] \geq 0. \tag{8.6}$$

*Proof.* Replace $\Phi$ by $\Phi^\sharp = \Phi + |Y - Y_0|^4 + |t - t_0|^4$. This makes the local maximum strict while preserving the derivatives through order two at the contact. Choose a small closed product neighborhood of $((y_i^0, t_0))_{i=1}^N$, contained in the domain, such that the diagonal points $(Y, t)$ lie in this strict neighborhood. For independent time variables $T = (t_1, \dots, t_N)$, put

$$\bar{t} = \frac{1}{N}\sum_i t_i, \qquad \chi(T) = \frac{1}{2}\sum_i (t_i - \bar{t})^2, \qquad \Psi_\lambda(Y, T) = \Phi^\sharp(Y, \bar{t}) + \lambda\chi(T).$$

Maximize $\sum_i \theta_i v(y_i, t_i) - \Psi_\lambda(Y, T)$ on the closed product. Comparison with the original diagonal point and boundedness on this product show $\lambda\chi(T_\lambda) \leq C$ at any maximum. Thus all time deviations tend to zero as $\lambda \to \infty$. Every limit of such maximum points is diagonal; dropping the nonnegative penalty in the maximum inequality shows that its diagonal limit is a maximum of $V - \Phi^\sharp$. Strictness forces this limit to be $(Y_0, t_0)$. Hence all the maximum points converge to the original point and are interior for sufficiently large $\lambda$.

Fix one such $\lambda$ and one maximum. Set

$$q_i = \theta_i^{-1}\nabla_{y_i}\Phi^\sharp(Y_\lambda, \bar{t}_\lambda), \qquad \zeta_\lambda = ((q_i, 1))_{i=1}^N, \qquad A_\lambda = D^2\Psi_\lambda(Y_\lambda, T_\lambda),$$

where $A_\lambda$ is written in the separate variables $((y_i, t_i))_i$. Apply Lemma 3.1 to the functions $\theta_i v(y_i, t_i)$ and the test $\Psi_\lambda$. For every $\eta > 0$ it supplies closed superjets with Hessians $X_i$ and first derivatives $D_{(y_i,t_i)}\Psi_\lambda$, with $\operatorname{diag}(X_i) \leq A_\lambda + \eta A_\lambda^2$. Dividing each jet by the positive number $\theta_i$ gives a closed superjet of $v$ with horizontal gradient $q_i$. The viscosity inequalities therefore give $X_i[(q_i, 1)] \geq 0$. Evaluating the matrix inequality on $\zeta_\lambda$ yields

$$0 \leq \sum_i X_i[(q_i, 1)] \leq A_\lambda[\zeta_\lambda] + \eta|A_\lambda\zeta_\lambda|^2.$$

First let $\eta \downarrow 0$, keeping $\lambda$ and its maximum fixed. The quantities $A_\lambda$ and $\zeta_\lambda$ do not depend on $\eta$, so $A_\lambda[\zeta_\lambda] \geq 0$. The time component of $\zeta_\lambda$ is $(1, \dots, 1)$, which is in the kernel of $D^2\chi$. The linear map $(Y, T) \mapsto (Y, \bar{t})$ sends $\zeta_\lambda$ to $((q_i)_i, 1)$. We conclude that

$$D^2\Phi^\sharp(Y_\lambda, \bar{t}_\lambda)[((q_i)_i, 1)] \geq 0.$$

Now let $\lambda \to \infty$. The maximum points converge to the contact, and the horizontal derivatives converge there. The derivatives of $\Phi^\sharp$ at that point are those of $\Phi$, giving (8.6). The time components of the first derivatives of $\Psi_\lambda$ need no bound: they do not enter the direction $(\nabla_x v, 1)$ in the individual viscosity inequalities. □

**Corollary 8.3** (A barycentric subsolution)**.** *Let $f \in C(\mathbb{R}^m \times (0,\infty))$ be a viscosity subsolution of* (1.2) *with subquadratic spatial growth on compact time intervals. Then $(x,t) \mapsto \mathcal{C}_t f(x,t)$ is a viscosity subsolution of* (1.2) *on $\mathbb{R}^m \times (0,\infty)$. For every $c \in \mathbb{R}$, the same holds for $(x,t) \mapsto \mathcal{C}_t f(x,t) - ct$.*

*Proof.* Continuity follows from Lemma 8.1. Fix $c \in \mathbb{R}$, and let $\phi$ touch $(x,t) \mapsto \mathcal{C}_t f(x,t) - ct$ from above at $(x_0,t_0)$ with $t_0 > 0$. Choose an attaining finite configuration $(\theta_i, y_i^0)$ for $\mathcal{C}_{t_0} f(x_0,t_0)$, and keep its positive weights fixed. For variable $Y = (y_i)_i$ let $X(Y) = \sum_i \theta_i y_i$ and define

$$\Phi(Y,t) = \phi(X(Y),t) + ct + \frac{\sum_i \theta_i |y_i - X(Y)|^2}{2t}.$$

The defining supremum for $\mathcal{C}_t$ shows that $\sum_i \theta_i f(y_i,t) - \Phi(Y,t)$ has a local maximum at $(Y_0,t_0)$: for nearby $(Y,t)$ it is at most $\mathcal{C}_t f(X(Y),t) - ct - \phi(X(Y),t)$, and equality holds at the chosen configuration. Set $p = \nabla_x \phi(x_0,t_0)$. The individual horizontal directions in Lemma 8.2 are

$$q_i = \theta_i^{-1} \nabla_{y_i} \Phi = p + \frac{y_i^0 - x_0}{t_0}.$$

Their weighted mean is $p$. Along the straight motion with velocities $y_i' = q_i$ and $t' = 1$, the mean has velocity $p$ and each offset $y_i - X$ has velocity $(y_i^0 - x_0)/t_0$. Consequently, along this motion the variance term is

$$\frac{\sum_i \theta_i |(y_i^0 - x_0)(1 + h/t_0)|^2}{2(t_0+h)} = \frac{t_0+h}{2t_0^2} \sum_i \theta_i |y_i^0 - x_0|^2,$$

which is affine in $h$. The term $ct$ is also affine, and the Hessian of the remaining term along this motion is $D^2\phi[(p,1)]$. Lemma 8.2, applied to the subsolution inequality for $f$, therefore gives $F[\phi](x_0,t_0) \geq 0$. This proves the claim for every $c$, including $c = 0$. □

## 8.3 Construction and slope of the contact

**Proposition 8.4** (Extremal envelope contact)**.** *If $\mathcal{A}_{m,C_0}$ contains a nonaffine member, then there exist $\kappa \in (0,\infty)$ and $v \in \mathcal{A}_{m,C_0}$ centered at $z_* = (0,1)$ for which* (8.2) *holds with $f = v$ and $I = \mathbb{R}$. For $t > 0$ set*

$$W(x,t) = \mathcal{C}_t v(x,t) - \frac{\kappa^2 t}{2}. \tag{8.7}$$

*This function is continuous, has subquadratic spatial growth on compact positive-time intervals, and is a viscosity subsolution of* (1.2) *on $\mathbb{R}^m \times (0,\infty)$. It satisfies*

$$v(x,t) - \frac{\kappa^2 t}{2} \leq W(x,t) \leq v(x,t) \quad (t > 0), \qquad W(z_*) = v(z_*) = 0, \tag{8.8}$$

*and $W(\cdot,t) - |x|^2/(2t)$ is concave for every $t > 0$. At every point of contact with $t > 0$, the barycentric gap satisfies*

$$W(x,t) = v(x,t) \quad \Longrightarrow \quad \frac{\mathcal{C}_t v(x,t) - v(x,t)}{t} = \frac{\kappa^2}{2} > 0. \tag{8.9}$$

*Finally,*

$$H_v(z_*) \in \mathbb{R}, \qquad \lim_{\ell \downarrow 0} \mathcal{S}_\ell^+ W(z_*) = H_v(z_*). \tag{8.10}$$

*Proof. A positive finite horizontal defect.* Start with a nonaffine member $v^{(0)}$ and center it at the origin using Lemma 7.1. Some horizontal slice of $v^{(0)}$ is nonaffine. To verify this, suppose instead that $v^{(0)}(x,t) = p(t) \cdot x + c(t)$ for every $t$. The coefficients $p, c$ are continuous, as

follows by evaluating the function at 0 and at the coordinate vectors. Direct optimization gives

$$\mathcal{P}_\ell v^{(0)}(x,t) = p(t+\ell)\cdot x + c(t+\ell) + \frac{\ell}{2}|p(t+\ell)|^2.$$

For every fixed $x, t$ this is convex in $\ell$. In each convexity inequality, the coefficient of $x$ must vanish, since the inequality holds for all $x \in \mathbb{R}^m$. Hence $p$ satisfies the affine interpolation identity on every interval, and so is affine in $t$. On the other hand, (7.2), applied at $x = \pm Ke$ and $|t| \le K$, implies $|p(t)| \le CL(K)^2$. An affine function of $t$ with this bound as $K \to \infty$ is constant. For constant $p$, the displayed forward formula and its backward counterpart

$$\mathcal{M}_\ell v^{(0)}(x,t) = p\cdot x + c(t-\ell) - \frac{\ell}{2}|p|^2$$

show that $c$ is both convex and concave. It is therefore affine, contradicting the choice of $v^{(0)}$.

A continuous nonaffine spatial slice has a nonzero midpoint Jensen defect, since the midpoint identity everywhere would make that slice affine. The transformation $f(x,t) \mapsto -f(x,-t)$ preserves $\mathcal{A}_{m,C_0}$: it exchanges the two viscosity inequalities, and it sends a fit $(p,a)$ to $(-p,a)$. It also changes the sign of a horizontal Jensen defect. Apply this transformation if needed, so that $v^{(0)}$ has a positive defect. Define

$$\kappa = \sup_{\substack{t,\ \text{finite configurations} \\ r>0}} \frac{\sum_i \theta_i v^{(0)}(y_i,t) - v^{(0)}(x,t)}{r}. \tag{8.11}$$

Then $\kappa > 0$. It is finite as well. For any configuration in this supremum, choose a fit at its mean $(x,t)$ of radius $r$ and use (7.1) with these data. At time zero the new positions are $(y_i - x)/r$, with mean zero and second moment one. The defect divided by $r$ is unchanged, since the affine terms cancel in the weighted difference. The new function is centered. Its growth bound implies $|\bar{v}(y,0)| \le C(1+|y|^2)$, with one constant for all these normalizations. Integrating this inequality over a configuration of second moment one bounds the quotient by a fixed constant. Thus $0 < \kappa < \infty$. More generally, (7.1) preserves every such quotient, at every time: spatial differences and their root mean square radius both scale by $R$, and the subtracted affine terms cancel.

*Attainment after compactness.* Choose configurations whose quotients in (8.11) tend to $\kappa$, and for each one make the normalization just described. Denote the resulting functions by $v_j$ and the normalized finite probability measures by $\mu_j$. Then

$$v_j(0,0) = 0, \qquad \int y\, d\mu_j = 0, \qquad \int |y|^2\, d\mu_j = 1, \qquad \int v_j(y,0)\, d\mu_j \longrightarrow \kappa.$$

Every $v_j$ is centered at the origin, and every finite configuration for it satisfies the Jensen bound with constant $\kappa$. First use Lemma 7.2 to pass to a locally uniform limit $\widehat{v} \in \mathcal{A}_{m,C_0}$, centered at the origin. The Jensen bound passes to this limit for each fixed finite configuration.

The moment bound gives $\mu_j(\{|y| > M\}) \le M^{-2}$, so the measures are tight. After a further subsequence they converge weakly to a probability measure $\mu$. Lower semicontinuity of the second moment gives $\int |y|^2\, d\mu \le 1$. The mean remains zero: the bound $\int_{|y|>M} |y|\, d\mu_j \le M^{-1}$ makes the first moments uniformly integrable, and the same bound holds for the limit. To retain the extremal defect, we also need convergence of the value integrals despite the varying functions and measures. For a compactly supported continuous cutoff $\chi_M$, local uniform convergence and weak convergence give

$$\int \chi_M(y) v_j(y,0)\, d\mu_j \longrightarrow \int \chi_M(y) \widehat{v}(y,0)\, d\mu.$$

Choose $\chi_M = 1$ on $\{|y| \le M\}$ and zero outside $\{|y| < 2M\}$. By (7.6),

$$\int_{|y|>M} |v_j(y,0)|\, d\mu_j \le \sup_{\substack{j \\ |y|>M}} \frac{|v_j(y,0)|}{|y|^2} \int |y|^2\, d\mu_j \longrightarrow 0$$

uniformly in $j$; the analogous tail bound holds for $\widehat{v}$ and $\mu$. Letting first $j \to \infty$ and then $M \to \infty$ proves

$$\int y\, d\mu = 0, \qquad \int |y|^2\, d\mu \le 1, \qquad \int \widehat{v}(y,0)\, d\mu - \widehat{v}(0,0) = \kappa. \tag{8.12}$$

Apply (8.3) to $\widehat{v}$. By Lemma 8.1, the measure in (8.12) is admissible for the value of $\mathcal{C}_{1/\kappa}\widehat{v}(0,0)$. It gives

$$\mathcal{C}_{1/\kappa}\widehat{v}(0,0) - \widehat{v}(0,0) \ge \kappa - \frac{\kappa}{2}\int |y|^2\, d\mu \ge \frac{\kappa}{2}.$$

This is equality by (8.3). In particular, loss of second moment in the weak limit causes no loss of contact; the equality also forces $\int |y|^2\, d\mu = 1$.

Set $R = 1/\kappa$, and choose a fit $(p_0, a_0)$ for $\widehat{v}$ at $(0,0)$ of radius $R$. Define the final function by the single change of frame and time translation

$$v(y,t) = \frac{\widehat{v}(R(y + p_0(t-1)), R(t-1)) - \widehat{v}(0,0)}{R} - p_0 \cdot y - \left(a_0 + \frac{|p_0|^2}{2}\right)(t-1).$$

It is centered at $(0,1)$ and still has the global Jensen bound with constant $\kappa$. Under the change of frame (7.1), at corresponding points $Z$ and $(y,s)$ the envelope gaps obey

$$\mathcal{C}_\sigma \bar{v}(y,s) - \bar{v}(y,s) = \frac{\mathcal{C}_{R\sigma}\widehat{v}(Z) - \widehat{v}(Z)}{R}.$$

This follows by scaling the spatial variance by $R^2$ in (8.1); the affine terms again cancel. The preceding equality at parameter $1/\kappa = R$ therefore becomes

$$\mathcal{C}_1 v(0,1) - v(0,1) = \frac{\kappa^2}{2}. \tag{8.13}$$

*The subsolution below $v$.* Define $W$ by (8.7). Equations (8.3) and (8.13) give (8.8). At any contact, (8.7) also gives (8.9). Lemma 8.1 gives continuity and the stated spatial semiconcavity. The sandwich and (7.6) give subquadratic spatial growth on every compact positive-time interval. Since $v$ is a solution with the growth in (7.6), Corollary 8.3, with $f = v$ and $c = \kappa^2/2$, gives the subsolution property of $W$.

*A finite slope at the contact.* Lemma 7.4 already bounds $H_v$ above at every point. We first bound it below at the finitely many points supporting the extremal configuration, using one common forward paraboloid, and then transfer this bound to their barycenter through equality in the Jensen bound. Choose a finite attaining configuration in (8.13), with mean $x = 0$, radius $r$, and positive weights. If $D = \sum_i \theta_i v(y_i, 1) - v(0,1)$, its attainment and the Jensen bound imply

$$\frac{\kappa^2}{2} = D - \frac{r^2}{2} \le \kappa r - \frac{r^2}{2} = \frac{\kappa^2}{2} - \frac{(r-\kappa)^2}{2}.$$

All inequalities are equalities, so

$$r = \kappa, \qquad \sum_i \theta_i v(y_i, 1) - v(0,1) = \kappa r. \tag{8.14}$$

Let $A$ be a minimizing slope in (8.4) for $(x,t,s)=(0,1,1)$. The weighted attaining value equals its affine majorant. Since all weights are positive, each $y_i$ must attain that majorant. The identity

$$v(y,1)-\frac{|y|^2}{2}-A\cdot y=v(y,1)-\frac{|y+A|^2}{2}+\frac{|A|^2}{2}$$

therefore shows that all $y_i$ are optimizers for $\mathcal{P}_1 v(b,0)$ with $b=-A$.

Put $h(s)=\mathcal{P}_s v(b,0)$ for $0\le s\le 1$. It is finite and convex by Lemma 7.3. For $0<\ell<1$, use the point $y^-_{i,\ell}=b+(1-\ell)(y_i-b)$ at time $1-\ell$ as a competitor for $\mathcal{M}_\ell v(y_i,1)$. Its value is bounded above using $\mathcal{P}_{1-\ell}v(b,0)$. The two quadratic costs on this linear path add, giving explicitly

$$\mathcal{M}_\ell v(y_i,1)\le h(1-\ell)+\frac{|y_i-b|^2}{2},\qquad v(y_i,1)=h(1)+\frac{|y_i-b|^2}{2}.$$

It follows that

$$\mathcal{S}^-_\ell v(y_i,1)\ge\frac{h(1)-h(1-\ell)}{\ell}\ge h(1)-h(0).$$

The last inequality is the chord inequality for the convex function $h$. Taking the slope limit shows $H_v(y_i,1)\ge h(1)-h(0)>-\infty$. These finitely many values are also finite above by Lemma 7.4.

For each $\ell>0$, choose an optimizer $\widehat{y}_i$ for $\mathcal{P}_\ell v(y_i,1)$ and let $\widehat{x}=\sum_i\theta_i\widehat{y}_i$. Write $\xi_i=y_i-x$, $\widehat{\xi}_i=\widehat{y}_i-\widehat{x}$, and $r'=(\sum_i\theta_i|\widehat{\xi}_i|^2)^{1/2}$. The exact orthogonal decomposition and the reverse triangle inequality in the weighted Euclidean space give

$$\sum_i\theta_i|\widehat{y}_i-y_i|^2=|\widehat{x}-x|^2+\sum_i\theta_i|\widehat{\xi}_i-\xi_i|^2\ge|\widehat{x}-x|^2+(r'-r)^2.$$

Use the Jensen bound at time $1+\ell$, (8.14) at time 1, and $\widehat{x}$ as a forward competitor from $x$. The decomposition yields

$$\begin{aligned}\ell\sum_i\theta_i\mathcal{S}^+_\ell v(y_i,1)&\le\ell\mathcal{S}^+_\ell v(x,1)+\kappa(r'-r)-\frac{(r'-r)^2}{2\ell}\\&\le\ell\mathcal{S}^+_\ell v(x,1)+\frac{\kappa^2\ell}{2}.\end{aligned}$$

Divide by $\ell$ and take $\ell\downarrow 0$. There are only finitely many endpoints, all with finite limits, so

$$H_v(z_*)\ge\sum_i\theta_i H_v(y_i,1)-\frac{\kappa^2}{2}\ge h(1)-h(0)-\frac{\kappa^2}{2}>-\infty.$$

Its upper finiteness is part of Lemma 7.4; hence $H_v(z_*)\in\mathbb{R}$.

It remains to identify the forward slope of $W$. At the contact, the sandwich gives for every $\ell>0$ $\mathcal{S}^+_\ell W(z_*)\le\mathcal{S}^+_\ell v(z_*)$. For $0<\ell<1$ it also gives $\mathcal{S}^-_\ell W(z_*)\ge\mathcal{S}^-_\ell v(z_*)$, since the backward time is positive. The subsolution property and growth of $W$ allow the one-sided inequality (7.12) to be applied to it. Consequently

$$H_v(z_*)\le\limsup_{\ell\downarrow0}\mathcal{S}^-_\ell W(z_*)\le\lim_{\ell\downarrow0}\mathcal{S}^+_\ell W(z_*)\le H_v(z_*).$$

This proves (8.10). □

# 9 Discrete propagation inequalities

We now derive the one-step inequality used to propagate the contact from Proposition 8.4. The argument will also be applied after normalizing at fitted points that need not be contacts, so the estimates must hold without assuming contact at the center. The walk constructed here uses forward and backward quadratic optimizers for the limiting equation; its drift controls velocity discrepancies that persist under fitted changes of frame.

Throughout this section, let $v \in \mathcal{A}_{m,C_0}$ be centered at $(0,1)$, and fix $0 \le \kappa < \infty$ for which (8.2) holds with $f = v$ and $I = \mathbb{R}$. For $t > 0$, define

$$W(x,t) = \mathcal{C}_t v(x,t) - \frac{\kappa^2 t}{2}.$$

The general bound (8.3), with its parameter equal to $t$, gives

$$v(x,t) - \frac{\kappa^2 t}{2} \le W(x,t) \le v(x,t).$$

Lemma 8.1 gives continuity of $W$ and concavity of $W(\cdot,t) - |\cdot|^2/(2t)$. The sandwich and (7.6) give subquadratic spatial growth for $W$ on compact positive-time intervals. Since $v$ is a subsolution with that growth, Corollary 8.3, applied with $c = \kappa^2/2$, makes $W$ a subsolution of (1.2).

For $\ell, \epsilon > 0$, set

$$f = W, \qquad f^\ell = \mathcal{P}_\ell f, \qquad g_\epsilon = \mathcal{M}_\epsilon v, \qquad J_\epsilon = \frac{f^\epsilon - g_\epsilon}{2\epsilon}. \tag{9.1}$$

These functions are defined at every positive time. The identity

$$\begin{aligned} 2\epsilon J_\epsilon(z) = W(z) - v(z) + \big(\mathcal{P}_\epsilon W(z) - W(z)\big) \\ + \big(v(z) - \mathcal{M}_\epsilon v(z)\big) \end{aligned} \tag{9.2}$$

and the zero-length continuity in Lemma 7.3 show that $2\epsilon J_\epsilon \to W - v$ uniformly on every compact subset of $\mathbb{R}^m \times (0,\infty)$. Thus, on a compact region separated from contact, $J_\epsilon$ tends uniformly to $-\infty$. This fact will turn lower expectation bounds for $J_\epsilon$ into contact at limiting marks. The sandwich also gives $\mathcal{P}_\epsilon W \le \mathcal{P}_\epsilon v$, and hence

$$J_\epsilon(z) \le \frac{1}{2}\big(\mathcal{S}_\epsilon^+ v(z) + \mathcal{S}_\epsilon^- v(z)\big). \tag{9.3}$$

For the particular pair supplied by Proposition 8.4, write $z_* = (0,1)$ and $H_* = H_v(z_*) \in \mathbb{R}$. Equality of $W$ and $v$ there, the contact slope conclusion, and the common slope limit for $v$ give

$$J_\epsilon(z_*) = \frac{1}{2}\big(\mathcal{S}_\epsilon^+ W(z_*) + \mathcal{S}_\epsilon^- v(z_*)\big) \longrightarrow H_* \qquad (\epsilon \downarrow 0). \tag{9.4}$$

This finite limiting value will start the walks in Section 10.

## 9.1 Spatial regularity and the one-step drift

**Lemma 9.1.** *For every $t, \ell > 0$, the optimization defining $f^\ell(x,t)$ has a unique maximizer $y = y(x,t,\ell)$. The function $f^\ell(\cdot,t)$ is differentiable, and*

$$\nabla_x f^\ell(x,t) = \frac{y - x}{\ell}. \tag{9.5}$$

*For every $h \in \mathbb{R}^m$ it satisfies the two supporting inequalities*

$$\begin{aligned} f^\ell(x,t) + \nabla_x f^\ell(x,t)\cdot h - \frac{|h|^2}{2\ell} &\le f^\ell(x+h,t) \\ &\le f^\ell(x,t) + \nabla_x f^\ell(x,t)\cdot h + \frac{|h|^2}{2t}. \end{aligned} \tag{9.6}$$

*In particular, its spatial Hessian is bounded below by $-\ell^{-1}I$ in the semiconvexity sense and above by $t^{-1}I$ in the semiconcavity sense.*

*Proof.* Put $s = t + \ell$. The function $f(\cdot, s) - |\cdot|^2/(2s)$ is concave, because the same is true of $\mathcal{C}_s v(\cdot, s) - |\cdot|^2/(2s)$. Thus

$$y \longmapsto f(y,s) - \frac{|y-x|^2}{2\ell}$$

is strictly concave: after subtracting the concave function $f(y,s) - |y|^2/(2s)$, its quadratic part has Hessian $(s^{-1} - \ell^{-1})I = -t(\ell s)^{-1}I$. It has a maximizer by the subquadratic growth and the quadratic cost, and strict concavity makes the maximizer unique. Keeping this maximizer fixed when $x$ is replaced by $x + h$ gives

$$f^\ell(x+h,t) \ge f^\ell(x,t) + \frac{y-x}{\ell}\cdot h - \frac{|h|^2}{2\ell}. \tag{9.7}$$

For the upper bound, the identity

$$\frac{|y|^2}{2s} - \frac{|y-x|^2}{2\ell} - \frac{|x|^2}{2t} = -\frac{t}{2\ell s}\left|y - \frac{s}{t}x\right|^2$$

shows that

$$f^\ell(x,t) - \frac{|x|^2}{2t} = \sup_y \left\{ f(y,s) - \frac{|y|^2}{2s} - \frac{t}{2\ell s}\left|y - \frac{s}{t}x\right|^2 \right\}$$

is concave in $x$. Indeed, the expression in braces is jointly concave in $(x, y)$; applying that concavity to a pair of maximizing points proves concavity of its partial supremum. A finite concave function on $\mathbb{R}^m$ has an upper supporting affine function at each point. Adding back the quadratic therefore gives a vector $q$ such that

$$f^\ell(x+h,t) \le f^\ell(x,t) + q\cdot h + \frac{|h|^2}{2t} \quad \text{for every } h.$$

Comparison with (9.7), first for $h = \rho e$ and then for $h = -\rho e$, and passage to $\rho \downarrow 0$, shows that $q = (y-x)/\ell$. The two quadratic remainders now prove differentiability and (9.5)–(9.6). □

Fix a state $z = (x,t)$ with $t > \epsilon$. Choose a maximizer for $\mathcal{P}_{2\epsilon} f(z)$ and a minimizer for $\mathcal{M}_{2\epsilon} v(z)$, and take $x^+$ and $x^-$ to be the respective intermediate points at time $t+\epsilon$ and $t-\epsilon$ on the straight paths to those optimizers. The composition formulas for the envelopes and additivity of the quadratic cost along a straight path give the following identities. Each half of a straight path has the same velocity and cost $\epsilon|Q_\pm|^2/2$.

$$\begin{aligned} f^{2\epsilon}(z) &= f^\epsilon(z^+) - \frac{\epsilon}{2}|Q_+|^2, \quad z^+ = (x^+, t+\epsilon), \\ \mathcal{M}_{2\epsilon} v(z) &= g_\epsilon(z^-) + \frac{\epsilon}{2}|Q_-|^2, \quad z^- = (x^-, t-\epsilon), \end{aligned} \tag{9.8}$$

where

$$Q_+ = \frac{x^+ - x}{\epsilon} = \nabla_x f^{2\epsilon}(z), \qquad Q_- = \frac{x - x^-}{\epsilon}, \qquad Q_0 = \nabla_x f^\epsilon(z). \tag{9.9}$$

The formula for $Q_+$ follows from (9.5) for the full $2\epsilon$ optimizer. Although $z^-$ is the backward step, $Q_-$ is oriented forward from $x^-$ to $x$; the spatial increment for that step is $-\epsilon Q_-$. The backward optimizer need not be unique; any choice has the properties just stated.

**Lemma 9.2.** *At the state and steps above define*

$$\begin{aligned} D &= f(z) + f^{2\epsilon}(z) - 2f^{\epsilon}(z), \\ e &= f^{\epsilon}(z^-) + \frac{\epsilon}{2}|Q_-|^2 - f(z), \\ \widetilde{D} &= 2g_{\epsilon}(z) - v(z) - \mathcal{M}_{2\epsilon}v(z), \\ \widetilde{e} &= v(z) + \frac{\epsilon}{2}|Q_+|^2 - g_{\epsilon}(z^+). \end{aligned}$$

*Here $D$ and $\widetilde{D}$ are the midpoint gaps for forward convexity and backward concavity. The quantities $e$ and $\widetilde{e}$ are the gaps from using $z$ as a competitor from the opposite step. All four quantities are nonnegative, and the exact drift identity and the velocity estimate are*

$$\begin{aligned} \frac{J_{\epsilon}(z^+) + J_{\epsilon}(z^-)}{2} - J_{\epsilon}(z) &= \frac{D + e + \widetilde{D} + \widetilde{e}}{4\epsilon} \\ &\geq \frac{D + e}{4\epsilon} \\ &\geq c(|Q_+ - Q_0|^2 + |Q_- - Q_0|^2), \qquad c = \frac{1}{16(1 + \log 2)}. \end{aligned} \tag{9.10}$$

For a fair choice between $z^+$ and $z^-$, the left side of (9.10) is the conditional expected change of $J_{\epsilon}$ given this state and the chosen optimizers. The drift therefore controls the squared discrepancies from the same reference velocity $Q_0$ at that state.

*Proof.* Convexity in the length of $\mathcal{P}_{\ell}f(z)$ gives $D \geq 0$, and concavity in the length of $\mathcal{M}_{\ell}v(z)$ gives $\widetilde{D} \geq 0$. The point $z$ is a forward competitor from $z^-$ and a backward competitor from $z^+$, which gives $e \geq 0$ and $\widetilde{e} \geq 0$. The identities in (9.8) and the definitions of the competitor gaps yield

$$\begin{aligned} f^{\epsilon}(z^+) + f^{\epsilon}(z^-) - 2f^{\epsilon}(z) &= D + e + \frac{\epsilon}{2}(|Q_+|^2 - |Q_-|^2), \\ g_{\epsilon}(z^+) + g_{\epsilon}(z^-) - 2g_{\epsilon}(z) &= -\widetilde{D} - \widetilde{e} + \frac{\epsilon}{2}(|Q_+|^2 - |Q_-|^2). \end{aligned}$$

The same difference of the two half-step costs occurs in both lines, so it cancels on subtraction. Dividing by $4\epsilon$ proves the identity and its first inequality.

For the final inequality let $b, d \in \mathbb{R}^m$ be arbitrary. Apply the path-convexity property of the forward envelope to

$$x(\ell) = x + d + \frac{\ell}{2\epsilon}(b - d), \qquad 0 \leq \ell \leq 2\epsilon.$$

For this path the function $\mathcal{P}_{\ell}f(x(\ell), t) + |b-d|^2\ell\log\ell/(4\epsilon^2)$ is convex, with $\ell\log\ell$ assigned its continuous value zero at $\ell = 0$. The midpoint inequality, and $2\epsilon\log(2\epsilon) - 2\epsilon\log\epsilon = 2\epsilon\log 2$, give

$$f(x + d, t) + f^{2\epsilon}(x + b, t) - 2f^{\epsilon}\left(x + \frac{b + d}{2}, t\right) \geq -\frac{\log 2}{2\epsilon}|b - d|^2. \tag{9.11}$$

Using $x + d$ as a forward competitor from $z^-$ and expanding its cost gives

$$f(x + d, t) \leq f(z) + e + Q_- \cdot d + \frac{|d|^2}{2\epsilon}.$$

Use the upper supporting inequality in (9.6) for $f^{2\epsilon}(x+b,t)$, and the lower one for $f^{\epsilon}(x+(b+d)/2,t)$. The left side of (9.11) is then at most

$$D+e+(Q_+-Q_0)\cdot b+(Q_--Q_0)\cdot d+\frac{|b|^2}{2t}+\frac{|d|^2}{2\epsilon}+\frac{|b+d|^2}{4\epsilon}.$$

Combining this upper bound with (9.11), using $t>\epsilon$ and $|b\pm d|^2\le 2(|b|^2+|d|^2)$, proves

$$D+e\ge (Q_0-Q_+)\cdot b+(Q_0-Q_-)\cdot d-\frac{1+\log 2}{\epsilon}(|b|^2+|d|^2).$$

Choose

$$b=\frac{\epsilon(Q_0-Q_+)}{2(1+\log 2)},\qquad d=\frac{\epsilon(Q_0-Q_-)}{2(1+\log 2)}.$$

The result is $D+e\ge \epsilon(|Q_+-Q_0|^2+|Q_--Q_0|^2)/(4(1+\log 2))$, as required. □

## 9.2 Fitted frames and a logarithmic velocity bound

The walks will later be viewed in frames based at fits at smaller positive times. We first show that the squared discrepancies in (9.10) are unchanged in those frames, and that the transformed solution remains in the standing family. The velocity estimate following this calculation will then apply with constants independent of the frame.

**Lemma 9.3** (Covariance in a fitted frame)**.** *Let $r>0$, and let $(p_0,a_0)$ be a fit for $v$ at $(x_0,r)$ of radius $r$. At corresponding points define*

$$\begin{aligned} s&=\frac{t}{r},\qquad y=\frac{x-x_0-p_0(t-r)}{r},\\ x&=x_0+r\big(y+p_0(s-1)\big),\qquad t=rs,\\ \bar v(y,s)&=\frac{v(x,t)-v(x_0,r)}{r}-p_0\cdot y-\left(a_0+\frac{|p_0|^2}{2}\right)(s-1). \end{aligned}\tag{9.12}$$

*Then $\bar v\in\mathcal{A}_{m,C_0}$ is centered at $(0,1)$ and satisfies (8.2) with $f=\bar v$, $I=\mathbb{R}$, and the same $\kappa$. Define*

$$\bar W(y,s)=\mathcal{C}_s\bar v(y,s)-\frac{\kappa^2 s}{2}\qquad (s>0).$$

*For every $\sigma>0$, the barycentric and $W$ gaps satisfy*

$$\begin{aligned} \mathcal{C}_\sigma\bar v(y,s)-\bar v(y,s)&=\frac{\mathcal{C}_{r\sigma}v(x,t)-v(x,t)}{r},\\ \bar W(y,s)-\bar v(y,s)&=\frac{W(x,t)-v(x,t)}{r}\qquad (s>0). \end{aligned}\tag{9.13}$$

*For every $\ell>0$, the solution slopes satisfy*

$$\mathcal{S}^{\pm}_{\ell}\bar v(y,s)=\mathcal{S}^{\pm}_{r\ell}v(x,t)-a_0.\tag{9.14}$$

*If the steps in (9.8) are chosen at any state with $t>\epsilon$, their images are optimizing steps for the transformed pair, with step size $\epsilon'=\epsilon/r$. Their velocities satisfy*

$$\begin{aligned} Q'_+&=\frac{y^+-y}{\epsilon'}=Q_+-p_0,\qquad Q'_-=\frac{y-y^-}{\epsilon'}=Q_--p_0,\\ Q'_0&=\nabla_y\mathcal{P}_{\epsilon'}\bar W(y,s)=Q_0-p_0,\qquad Q'_\pm-Q'_0=Q_\pm-Q_0. \end{aligned}\tag{9.15}$$

*All these identities are global at corresponding states, with the indicated positive-time restriction for $W$; the states need not lie in the fitting box. No contact at $(x_0,r)$ is assumed. The centering and fit assertion concerns $\bar v$ alone.*

*Proof.* Lemma 7.1, followed by the time translation that puts the fitted point at $(0,1)$, proves the class and centering assertions. At each fixed time the spatial offsets are multiplied by $r$, and values are multiplied by $r$ up to an affine function. The affine terms cancel from a barycentric defect, so its quotient by the root mean square radius is unchanged. This proves the Jensen assertion.

For the remaining calculations put

$$A(y,s) = \frac{v(x_0,r)}{r} + p_0 \cdot y + \left(a_0 + \frac{|p_0|^2}{2}\right)(s-1), \qquad \frac{v(x,t)}{r} = \bar{v}(y,s) + A(y,s).$$

Scaling the positions of a barycentric configuration and cancelling its weighted affine terms gives

$$\frac{\mathcal{C}_{r\sigma} v(x,t)}{r} = \mathcal{C}_\sigma \bar{v}(y,s) + A(y,s).$$

Subtracting the displayed formula for $v/r$ proves the first identity in (9.13). For $s > 0$, take $\sigma = s$. The old time and the old barycentric parameter in $W$ are both $t = rs$, while the correction $\kappa^2 t/(2r)$ is $\kappa^2 s/2$. Hence

$$\frac{W(x,t)}{r} = \bar{W}(y,s) + A(y,s),$$

which proves the second gap identity. This is why the frame uses $s = t/r$: it preserves the time-zero boundary in the definition of $W$.

The quadratic envelopes obey the following value identities for $\ell > 0$, with $s > 0$ in the identity involving $W$:

$$\begin{aligned} \frac{\mathcal{P}_{r\ell} v(x,t)}{r} &= \mathcal{P}_\ell \bar{v}(y,s) + A(y,s) + a_0 \ell, \\ \frac{\mathcal{M}_{r\ell} v(x,t)}{r} &= \mathcal{M}_\ell \bar{v}(y,s) + A(y,s) - a_0 \ell, \\ \frac{\mathcal{P}_{r\ell} W(x,t)}{r} &= \mathcal{P}_\ell \bar{W}(y,s) + A(y,s) + a_0 \ell. \end{aligned} \tag{9.16}$$

To check the first identity, parameterize an old forward competitor by a new position $y'$ at time $s + \ell$. Its old displacement divided by $r$ is $y' - y + p_0 \ell$. The cost divided by $r$ is

$$\frac{|y' - y + p_0 \ell|^2}{2\ell} = \frac{|y' - y|^2}{2\ell} + p_0 \cdot (y' - y) + \frac{|p_0|^2 \ell}{2}.$$

The affine change in $A$ from $(y,s)$ to $(y', s+\ell)$ is $p_0 \cdot (y' - y) + (a_0 + |p_0|^2/2)\ell$, leaving $a_0 \ell$ after the cost is subtracted. For a backward competitor the old displacement divided by $r$ is $y' - y - p_0 \ell$; the same expansion with the cost added leaves $-a_0 \ell$. These prove the first two identities. The last follows by the same forward calculation using $W/r = \bar{W} + A$ at every positive time. Subtract the formula for $v/r$ from the forward identity, and subtract the backward identity from that formula. Dividing each result by $\ell$ gives (9.14).

The competitor calculations also show that old optimizers map to optimizers for the transformed envelopes. The affine coordinate map sends their straight intermediate points to the new straight intermediate points. Thus the mapped steps are precisely the steps of (9.8) for the transformed pair. The first two velocity equalities in (9.15) follow from (9.12) and $t^\pm - t = \pm\epsilon$. For the reference velocity, differentiate the last identity of (9.16) in $y$ with $r\ell = \epsilon$. The derivative of the left side is $\nabla_x \mathcal{P}_\epsilon W = Q_0$, since $D_y x = rI$, and the derivative of $A$ is $p_0$. This proves the equality for $Q_0'$ and the invariance of the velocity discrepancies, with no additional scale factor. □

It remains to bound the reference velocity in each of these centered frames. As before, write $L(K) = 1 + \log K$ for $K \geq 2$. The next bound is uniform over the standing family. Its squared logarithmic singularity in time is integrable near zero, which will control the expected martingale variation through the time-walk occupation estimates.

**Lemma 9.4.** *Suppose $K \geq 2$, $|x|, |t| \leq K$, $0 < \epsilon \leq 1$, and $t > \epsilon$. Then the velocity in* (9.9) *satisfies*

$$|Q_0| = |\nabla_x \mathcal{P}_\epsilon W(x,t)| \leq C\big(L(K) + |\log t|\big). \tag{9.17}$$

*Here $C$ depends only on the fitting constant, the spatial dimension, and $\kappa$.*

*Proof.* Let $s = t + \epsilon$, so $2\epsilon < s \leq K + 1$. At the point $x$, choose any upper supporting slope $p$ for the semiconcave function $f(\cdot, s)$:

$$f(x+h,s) \leq f(x,s) + p \cdot h + \frac{|h|^2}{2s} \quad (h \in \mathbb{R}^m). \tag{9.18}$$

Such a slope exists because $f(\cdot,s) - |\cdot|^2/(2s)$ is finite and concave on all of $\mathbb{R}^m$. If $p \neq 0$, insert $h = -sp/|p|$ to obtain

$$f(x+h,s) - f(x,s) \leq -s|p| + \frac{s}{2}.$$

The sandwich bound for $W$ gives, in the other direction,

$$f(x+h,s) - f(x,s) \geq v(x+h,s) - v(x,s) - \frac{\kappa^2 s}{2}.$$

The spatial increment estimate from the fits, applied with increment length $s$ at the center $(x,s)$ in the box of size $2K$, bounds the right side below by

$$-Cs(L(2K) + |\log s|) - \frac{\kappa^2 s}{2}.$$

The resulting estimate for $p$ also holds when $p = 0$. Since $1 < s/t < 2$, we have $|\log s| \leq |\log t| + \log 2$, and hence

$$|p| \leq C\big(L(K) + |\log t|\big). \tag{9.19}$$

Let $h$ be the optimizing displacement for $f^\epsilon(x,t)$. Its value is at least the value of the competitor $h = 0$. In conjunction with (9.18), this gives

$$\frac{|h|^2}{2\epsilon} \leq p \cdot h + \frac{|h|^2}{2s}.$$

For $h \neq 0$, using $s > 2\epsilon$ yields $|h|/\epsilon \leq 4|p|$; the same inequality is immediate when $h = 0$. By (9.5), $Q_0 = h/\epsilon$, so (9.19) proves the lemma. □

# 10 Propagation to time zero and localization

We now apply the discrete inequalities to the extremal pair $v, W$ and the constant $\kappa$ supplied by Proposition 8.4. Write $z_* = (0,1)$ and $H_* = H_v(z_*) \in \mathbb{R}$; the starting limit is (9.4).

**Proposition 10.1.** *Put $\delta_j = 2^{-j}$ for $j \geq 1$. There exist a constant $K < \infty$, a point $x_{\mathrm{end}} \in \mathbb{R}^m$, and an infinite set $\mathcal{J} \subset \mathbb{N}$ with the following property. For every $j \in \mathcal{J}$ there are a point $x_j \in \mathbb{R}^m$ and a fit $(p_j, a_j)$ for $v$ at $(x_j, \delta_j)$ of radius $\delta_j$, in the sense of* (1.3)*, such that*

$$\begin{gathered} H_v(x_{\mathrm{end}}, 0) > -\infty, \qquad W(x_j, \delta_j) = v(x_j, \delta_j), \\ |x_{\mathrm{end}} - x_j + p_j \delta_j| \leq K\delta_j. \end{gathered} \tag{10.1}$$

The last bound says that the same terminal point stays in a fixed bounded part of every selected fitted frame, where its finite slope will identify the limiting fixed-length slopes.

*Proof.* We first construct finite stopped walks with common expectation and error bounds, and then localize their terminal points in the frames of their marks. Finally, one joint limiting law will retain the terminal point and all marks, allowing us to select infinitely many localized contacts for the same endpoint.

*The stopped walks and their expectation bounds.* Fix for the moment an integer $T \geq 2$ and a radius $R \geq 2$. For each $N \geq 1$ choose $\epsilon_N$ to be an inverse power of two with

$$0 < \epsilon_N \leq \frac{\delta_N}{4}. \tag{10.2}$$

In particular, $\epsilon_N$ divides $1, T$, and every $\delta_j$ with $j \leq N$. We will make each $\epsilon_N$ smaller if necessary, retaining these conditions. Write $\epsilon = \epsilon_N$ while constructing the $N$-th walk.

Start at $z_0 = z_*$. At a state $z_n = (x_n, t_n)$ with $\delta_N < t_n < T$ and $|x_n| < R$, choose the two steps of (9.8) and take $z_{n+1} = z_n^+$ or $z_n^-$ with equal probabilities. Stop on first reaching $t_n = \delta_N$, $t_n = T$, or $|x_n| \geq R$; call this stopping index $\tau_N$. The choices of optimizers can be made separately at every node of the countable tree of finite histories. Taking independent fair signs $\xi_n \in \{-1, 1\}$ to select the steps then defines the walk and the filtration $\mathcal{F}_n = \sigma(\xi_0, \ldots, \xi_{n-1})$; all velocities at step $n$ are $\mathcal{F}_n$-measurable. In expressions such as $Q_-(z_n)$ below, the velocity denotes the choice made at that history. Before stopping, $t_n > \delta_N \geq 4\epsilon$, so the inequality (9.10) applies.

We choose $\epsilon_N$ sufficiently small that every position up to and including the stop lies in $\overline{B}_{R+1}(0)$. Here is the uniform localization of optimizers that permits this choice. The sandwich for $W$ extends it continuously to $t = 0$ by $W(x, 0) = v(x, 0)$. The growth bounds for $v$ and the sandwich give uniform subquadratic growth for $v$ and $W$ on compact time intervals in $\mathbb{R}$ and $[0, \infty)$, respectively. For base points in a fixed compact box and lengths $\ell \leq 1$, a quadratic cost first confines all optimizing positions for $\mathcal{P}_\ell W$ and $\mathcal{M}_\ell v$ to a fixed spatial compact set: outside a sufficiently large ball the cost dominates the uniform subquadratic bound, even with the weaker coefficient corresponding to $\ell = 1$. On that compact set the functions are bounded. Comparing an optimizer with the zero displacement competitor then gives $|y - x|^2/(2\ell) \leq C$ there, uniformly as $\ell \downarrow 0$. Thus the optimizing displacements tend to zero uniformly. Apply this to length $2\epsilon$ and base points with $|x| \leq R$, $0 \leq t \leq T$, in a slightly larger time interval. The intermediate displacements used for the steps are half the full displacements, so they can all be made at most one. This proves the asserted bound at stopping. We also choose $\epsilon_N$ small enough that

$$J_{\epsilon_N}(z_*) \geq H_* - 1, \tag{10.3}$$

which is possible by (9.4).

The time coordinate before stopping is the simple symmetric walk with step $\epsilon$ starting at 1. It is stopped no later than this time walk hits $\delta_N$ or $T$. The latter hitting time is almost surely finite and has finite expectation, for example by the elementary gambler's ruin formula. The same is therefore true of $\tau_N$. Put

$$\mathcal{B} = \overline{B}_{R+1}(0) \times [0, T], \qquad B = R + T + 2.$$

The upper slope bound for $v$ on boxes and (9.3) give a constant $U_B = CL(B)^4$, independent of $N$, such that

$$J_{\epsilon_N}(z_n) \leq U_B \qquad (0 \leq n \leq \tau_N). \tag{10.4}$$

For any stopping time $\sigma \le \tau_N$, summing the conditional form of (9.10) up to $\sigma \wedge h$ gives

$$\mathbb{E}J_\epsilon(z_{\sigma\wedge h}) \ge J_\epsilon(z_*) + c\,\mathbb{E}\sum_{n<\sigma\wedge h}\bigl(|Q_+(z_n)-Q_0(z_n)|^2 + |Q_-(z_n)-Q_0(z_n)|^2\bigr). \tag{10.5}$$

For each fixed $N$, $J_\epsilon$ is continuous and bounded on the compact box $\overline{B}_{R+1}(0)\times[\delta_N,T]$ containing the stopped walk. Thus dominated convergence applies to its term as $h\to\infty$. Monotone convergence applies to the nonnegative sum. We obtain, for all these stopping times,

$$\mathbb{E}J_\epsilon(z_\sigma) \ge J_\epsilon(z_*). \tag{10.6}$$

Taking $\sigma=\tau_N$ in (10.5) and using (10.3)–(10.4) also gives

$$\mathbb{E}\sum_{n<\tau_N}\bigl(|Q_+(z_n)-Q_0(z_n)|^2 + |Q_-(z_n)-Q_0(z_n)|^2\bigr) \le B_0, \qquad B_0 = C_*L(B)^4. \tag{10.7}$$

Here $C_*$ is fixed for $v$ and independent of $N,R,T$; it absorbs the finite number $H_*$ and the absolute constant $c$ in (9.10). Both (10.6) and (10.7) are unconditional expectations. Together with the upper slope bound, the lower expectation bound will give contact and a finite terminal slope in the limit. We first use the discrepancy bound to control horizontal-exit probabilities, both in the original coordinates and in fitted frames.

*Reaching the lower level with positive probability.* We next show how to choose the outer cutoffs so that the walks reach $t=\delta_N$ with probability at least $3/4$, uniformly in $N$. Couple the time signs with the simple symmetric walk starting at 1 and stopped only at 0 or $T$. Its probability of reaching $T$ is $1/T$, which also bounds that probability for our walk, since it may stop sooner. To control horizontal exit, we will bound the expected maximal displacement by $C_TL(R+T+2)^2$, with $C_T$ independent of $R$ and $N$. Markov's inequality will then make the probability of horizontal exit tend to zero as $R\to\infty$ for fixed $T$.

The expected number of visits of the unrestricted time walk to any fixed interior grid level is at most $CT/\epsilon$. To verify this directly, write $q=T/\epsilon$ and identify the level with $k\in\{1,\dots,q-1\}$. After a visit to $k$, the probability of reaching an endpoint before the next return is

$$\frac{1}{2}\left(\frac{1}{k}+\frac{1}{q-k}\right)\ge\frac{2}{q}.$$

This follows from the linear hitting probability for a symmetric walk on each of the intervals $[0,k]$ and $[k,q]$. Starting from the first visit, the expected total number of visits is at most $q/2$, by the geometric return calculation; the probability of that first visit is at most one. Stopping the actual walk earlier can only decrease its visit counts. Summing this bound over levels gives

$$\mathbb{E}(\epsilon^2\tau_N)\le CT^2, \qquad \mathbb{E}\sum_{n<\tau_N}\epsilon^2|Q_0(z_n)|^2 \le C_TL(B)^2. \tag{10.8}$$

For the second inequality we used Lemma 9.4 before each step. More explicitly, it bounds the summand velocity squared by $C(L(B)^2+|\log t_n|^2)$, while the visit estimate gives at most

$$CT\epsilon\sum_{k=1}^{q-1}\bigl(L(B)^2+|\log(k\epsilon)|^2\bigr).$$

This has the asserted bound. Indeed, on $(0,1]$ the function $|\log s|^2$ is decreasing and integrable, so its right-endpoint grid sum times $\epsilon$ is at most its integral there; on $[1,T]$ it is bounded by $(\log T)^2$. The constants $C_T$ in this proof can depend on $T$ and the fixed data, but not on $R$ or $N$.

For $n < \tau_N$ the spatial increment can be written exactly as

$$x_{n+1} - x_n = \epsilon\xi_n Q_0(z_n) + \epsilon\xi_n\big(Q_{\xi_n}(z_n) - Q_0(z_n)\big),$$

where $Q_1 = Q_+$ and $Q_{-1} = Q_-$. Extend both kinds of increments by zero after stopping. The first summands are martingale differences, since the active-step indicator and $Q_0(z_n)$ are known before the fair sign. For a deterministic truncation in the number of steps, Doob's $L^2$ maximal inequality and orthogonality of these differences give an upper bound

$$2\left(\mathbb{E}\sum_{n<\tau_N}\epsilon^2|Q_0(z_n)|^2\right)^{1/2}$$

for the expected supremum of the norm of their partial sums. Letting the truncation increase preserves this bound by monotone convergence of the suprema. On each path, the sum of the norms of the other increments is at most

$$(\epsilon^2\tau_N)^{1/2}\left(\sum_{n<\tau_N}\big(|Q_+(z_n) - Q_0(z_n)|^2 + |Q_-(z_n) - Q_0(z_n)|^2\big)\right)^{1/2}.$$

Cauchy–Schwarz in expectation gives the travel inequality

$$\begin{aligned}\mathbb{E}\sup_{0\le n\le\tau_N}|x_n| \le 2&\left(\mathbb{E}\sum_{n<\tau_N}\epsilon^2|Q_0(z_n)|^2\right)^{1/2}\\ &+\Big(\mathbb{E}(\epsilon^2\tau_N)\Big)^{1/2}\\ &\times\left(\mathbb{E}\sum_{n<\tau_N}\big(|Q_+(z_n) - Q_0(z_n)|^2 + |Q_-(z_n) - Q_0(z_n)|^2\big)\right)^{1/2}.\end{aligned} \tag{10.9}$$

This calculation uses only predictability of the coefficients of the fair signs in the reference increments, including the active-step indicators, and the pathwise bound on error lengths. It therefore applies as well to the displacement from the starting point of a stopped portion after a stopping time, with its increments extended by zero outside that portion. Equations (10.7) and (10.8) now show that

$$\mathbb{E}\sup_{0\le n\le\tau_N}|x_n| \le C_T\big(L(B) + \sqrt{B_0}\big) \le C_T' L(B)^2. \tag{10.10}$$

The probability of a horizontal stop is at most the right side of (10.10) divided by $R$. For fixed $T$ this tends to zero as $R \to \infty$. Choose an integer $T \ge 8$, and then choose $R$ large enough that the horizontal-stop probability is at most $1/8$, uniformly in $N$. Fix these $T, R$ for the rest of the proof, so $B$ and $B_0$ are now fixed finite constants. If

$$\eta_N = \mathbf{1}_{\{t_{\tau_N} = \delta_N\}},$$

the two preceding bounds and the definition of the stop imply

$$\mathbb{P}(\eta_N = 1) \ge \frac{3}{4} \qquad (N \ge 1). \tag{10.11}$$

This remains true if the lower time level and the horizontal cutoff are reached simultaneously, since that case has $\eta_N = 1$.

*Marks and localization in fitted frames.* For $1 \le j \le N$, let

$$\sigma_{j,N} = \tau_N \wedge \inf\{n \ge 0 : t_n = \delta_j\}, \qquad z_j = (x_j, t_j) = z_{\sigma_{j,N}}.$$

Thus a mark is the first point on its time level, or the terminal point if the walk stops before that level. The divisibility of the grid ensures $t_j \geq \delta_j$, with equality on $\{\eta_N = 1\}$. At the time of this mark choose a fit $(p_j, a_j)$ at $z_j$ of radius $\delta_j$. These choices can again be made on the countable tree of histories, so the parameters are $\mathcal{F}_{\sigma_{j,N}}$-measurable. The bound on fit parameters on boxes gives, for a fixed constant independent of $N$,

$$|p_j| + |a_j| \leq \Gamma_j := C(L(B) + j), \qquad j \leq N,$$

increasing $C$ to make every $\Gamma_j \geq 1$.

The cutoffs $T, R$ and the error bound $B_0$ are now fixed. We will choose two further cutoffs, uniformly in $j, N$, to control the terminal point in each fitted frame. Fix $1 \leq j \leq N$, put $r = \delta_j$, and let $\mathcal{R}_{j,N} = \{t_j = r\}$ be the event that the level $r$ is reached, including the case where it is reached at the original stop. This event is $\mathcal{F}_{\sigma_{j,N}}$-measurable. On it $z_j = (x_j, r)$, so we may apply Lemma 9.3 pathwise with the chosen fit $(p_j, a_j)$. In its coordinates,

$$s = \frac{t}{r}, \qquad y = \frac{x - x_j - p_j(t - r)}{r}, \qquad \epsilon' = \frac{\epsilon_N}{r}.$$

The transformed solution $\bar{v}$ is centered at $(0, 1)$ and has the same Jensen bound. Its $\bar{W}$ therefore has the standing properties of Section 9. The mapped steps are optimizing steps, and (9.15) gives identical squared velocity discrepancies. Contact at this finite mark is not needed. The fit and frame are known at the marking time and will be held fixed afterward.

On $\mathcal{R}_{j,N}$ consider the portion of the original walk from its $j$-th mark to its stop, with the additional rule that this portion stops on first reaching $s = M$ or $|y| \geq K$, where $M \geq 2$ is an integer and $K \geq 2$. It includes the endpoint of this additional stopping rule. If the original walk stopped at the mark, the portion has no steps. A hit of an additional cutoff simultaneous with the original stop is counted as a hit of that additional cutoff. On $\mathcal{R}_{j,N}^{\mathrm{c}}$ define the portion to be empty as well, and assign zero to all its sums and spatial suprema. Let $n'$ be its number of steps, and write $\sum'$ for a sum over its pre-step states.

The total squared velocity error on this portion, in the new frame, has the unconditional bound

$$\mathbb{E} \sum{}' (|Q'_+ - Q'_0|^2 + |Q'_- - Q'_0|^2) \leq B_0. \tag{10.12}$$

Indeed, on each path the sum is a subsum of the original sum in (10.7), with identical summands by (9.15); on an empty portion it is zero. This argument gives an unconditional estimate. It makes no assertion that the same bound holds after conditioning on a particular history at the mark.

The time-walk estimates, in contrast, do hold conditionally on that history. On $\mathcal{R}_{j,N}$ the new time starts at $s = 1$ and has step $\epsilon' = \epsilon_N / r \leq 1/4$, which divides 1 and $M$. Given $\mathcal{F}_{\sigma_{j,N}}$, the subsequent signs remain fair. The portion is stopped no later than the unrestricted new time walk first hits 0 or $M$. Its conditional expected visits to each interior level are therefore at most $CM/\epsilon'$ by the same visit argument used above. Before any step of the portion, the original walk is still live, so $s = t/r > \delta_N / r \geq 4\epsilon'$. Also $|y| < K$ and $s < M$ before the additional stop. Lemma 9.4 applies to the centered $\bar{v}$ and its $\bar{W}$ uniformly, and bounds $|Q'_0|$ by $C(L(K + M + 2) + |\log s|)$. The integrable logarithmic grid sum gives

$$\begin{aligned} \mathbb{E}((\epsilon')^2 n' \mid \mathcal{F}_{\sigma_{j,N}}) &\leq CM^2 \mathbf{1}_{\mathcal{R}_{j,N}}, \\ \mathbb{E}\left( \sum{}' (\epsilon')^2 |Q'_0|^2 \,\middle|\, \mathcal{F}_{\sigma_{j,N}} \right) &\leq C_M L (K + M + 2)^2 \mathbf{1}_{\mathcal{R}_{j,N}}. \end{aligned} \tag{10.13}$$

The frame is known at the conditioning time, and the estimates of the centered class have constants independent of that frame. This explains the uniformity of these conditional bounds. Integrating them gives the same bounds unconditionally without the indicator.

Extend the new increments by zero outside the portion. Their reference part is $\epsilon'\xi_n Q'_0$ and their error part is $\epsilon'\xi_n(Q'_{\xi_n} - Q'_0)$. The reference increments are martingale differences after the random starting mark: the fit and reached event are known at that mark, and on an active step the indicator and $Q'_0 = Q_0 - p_j$ are $\mathcal{F}_n$-measurable while $\xi_n$ is still fair given $\mathcal{F}_n$. Thus the calculation of (10.9) applies in the primed variables. Using the integrated bounds from (10.13) and the unconditional bound (10.12), it gives

$$\mathbb{E}\left(\sup_{\text{points of the portion}} |y|\right) \le C'_M\big(L(K+M+2)+\sqrt{B_0}\big). \tag{10.14}$$

The supremum is understood to be zero for an empty portion. In particular, the probability that this portion reaches $|y| \ge K$ is at most the right side of (10.14) divided by $K$. The probability it reaches $s = M$ is at most $1/M$: conditionally on reaching the mark, the unrestricted symmetric time walk starting from 1 hits $M$ before 0 with probability $1/M$, and an earlier stop only removes this event.

Choose a fixed integer $M \ge 8$, and then choose $K$ so large that

$$\frac{C'_M\big(L(K+M+2)+\sqrt{B_0}\big)}{K} \le \frac{1}{8}.$$

This is possible because $B_0$ is already fixed and $L(K+M+2)/K \to 0$. On $\{\eta_N = 1\}$ the $j$-th level is reached. If neither additional cutoff intervenes, the terminal point is in the portion and has $|y| \le K$. Combining this observation with (10.11) and the two additional-cutoff probabilities proves, uniformly for $j \le N$,

$$\mathbb{P}\left(\eta_N = 1, \quad |x_{\tau_N} - x_j - p_j(t_{\tau_N} - \delta_j)| \le K\delta_j\right) \ge \frac{3}{4} - \frac{1}{8} - \frac{1}{8} = \frac{1}{2}. \tag{10.15}$$

*One limiting law and the contact properties.* We use one fixed compact space for all these laws. Define

$$\mathcal{K}_j = \mathcal{B} \times \overline{B}_{\Gamma_j}(0) \times [-\Gamma_j, \Gamma_j], \qquad \mathcal{X} = \mathcal{B} \times \{0,1\} \times \prod_{j=1}^{\infty} \mathcal{K}_j.$$

The coordinates are the terminal point, the flag, and the marked point and fit at each index. This countable product of compact metric spaces is compact and metrizable. For indices $j > N$, fill the coordinates by any fixed point of $\mathcal{K}_j$; these dummy entries need not be fits. Let $\mu_N$ be the law on $\mathcal{X}$ of the terminal point $z_{\tau_N}$, the flag $\eta_N$, and the marked points and fits. Compactness of the space of probability measures on a compact metric space gives a subsequence converging weakly to a probability law $\mu$; all subsequent limits in $N$ are along this subsequence. Denote its coordinates by $z_{\text{end}} = (x_{\text{end}}, t_{\text{end}})$, $\eta$, and $(z_j, p_j, a_j)$.

Here are first the properties that pass directly to this law. The flag is a coordinate in a discrete space, so its probability converges and $\mu(\eta = 1) \ge 3/4$. Also the continuous nonnegative function $\eta t_{\text{end}}$ on $\mathcal{X}$ has integral at most $\delta_N$ under $\mu_N$. Its integral under $\mu$ is zero. It follows that $t_{\text{end}} = 0$ almost surely on $\{\eta = 1\}$. For each fixed $j$, the sets $\{t_j \ge \delta_j\}$ and $\{\eta = 0\} \cup \{t_j = \delta_j\}$ are closed and have $\mu_N$-measure one for all $N \ge j$. The closed-set direction of the Portmanteau Theorem, $\limsup_N \mu_N(F) \le \mu(F)$, makes their $\mu$-measures one. Fits at the specified radius form a closed subset of $\mathcal{K}_j$. To see this without varying the sheared domain, write its increments in (1.3) as

$$\ell = \delta_j\theta, \qquad h = p_j\delta_j\theta + \delta_j\zeta, \qquad |\theta| \le 1, \quad |\zeta| \le 1.$$

For each $(\theta, \zeta)$ the fitting inequality is closed in $(z_j, p_j, a_j)$ by continuity of $v$, and their intersection over this fixed compact set of increments is closed. Portmanteau therefore also

passes the fitting property to $\mu$. Taking the intersection of the resulting full-measure sets for all $j$, we have

$$\begin{gathered} t_{\mathrm{end}} = 0 \text{ on } \{\eta = 1\}, \qquad t_j \geq \delta_j \text{ and } t_j = \delta_j \text{ on } \{\eta = 1\}, \\ (p_j, a_j) \text{ is a fit at } z_j \text{ of radius } \delta_j \quad \text{for every } j, \end{gathered} \tag{10.16}$$

almost surely.

Two further properties come from the expectation bounds:

$$H_v(z_{\mathrm{end}}) > -\infty, \qquad W(z_j) = v(z_j) \quad (j \geq 1) \qquad \text{almost surely under } \mu. \tag{10.17}$$

For the first, fix $\ell > 0$ and set

$$\Phi_\ell(z) = \frac{1}{2}(\mathcal{S}_\ell^+ v(z) + \mathcal{S}_\ell^- v(z)).$$

This is a finite continuous function on $\mathcal{B}$. For all sufficiently large $N$, $\epsilon_N \leq \ell$, and monotonicity of both slopes with respect to their length and (9.3) give $J_{\epsilon_N} \leq \Phi_\ell$. Apply (10.6) at $\tau_N$ and pass to the weak limit using this fixed continuous function and (9.4). The result is

$$\int \Phi_\ell(z_{\mathrm{end}})\, d\mu \geq H_* \qquad (\ell > 0). \tag{10.18}$$

Take a decreasing sequence $\ell_h \downarrow 0$ with $\ell_h \leq 1$. The functions $\Phi_{\ell_h}$ decrease to $H_v$ pointwise, because the two slope limits agree for $v$. Their uniform upper bound on $\mathcal{B}$ is $U_B$ after increasing its fixed constant. Hence $U_B - \Phi_{\ell_h}(z_{\mathrm{end}})$ are nonnegative and increase to $U_B - H_v(z_{\mathrm{end}})$. Monotone convergence and (10.18) give

$$\int \big(U_B - H_v(z_{\mathrm{end}})\big)\, d\mu \leq U_B - H_* < \infty.$$

The integrand would be infinite where $H_v = -\infty$, proving the first claim of (10.17).

For the contact claim fix $j$ and let $\nu_N^j$ and $\nu^j$ be the marginals of $\mu_N$ and $\mu$ for the marked point $z_j$. They converge weakly. Suppose that $\nu^j$ gives positive mass to $\{v - W > 0\}$. All these limit marks have $t \geq \delta_j$. Since $v - W$ is continuous and nonnegative for $t > 0$, there is $\alpha > 0$ such that the relatively open set

$$U = \{(x,t) \in \mathcal{B} : t > \delta_j/2,\ v(x,t) - W(x,t) > \alpha\}$$

has $\nu^j(U) > 0$ and its closure is contained in $\{t \geq \delta_j/2,\ v - W \geq \alpha\}$. For example, first choose $\alpha$ with positive mass where $v - W > 2\alpha$. The zero-length continuity of the quadratic envelopes is uniform on this compact closure. Thus (9.2) gives

$$2\epsilon_N J_{\epsilon_N}(z) \leq -\frac{\alpha}{2}$$

on $U$ for all sufficiently large $N$. In particular, $J_{\epsilon_N} \leq -\alpha/(4\epsilon_N)$ on $U$. By the open-set direction of Portmanteau, $\nu^j(U) \leq \liminf_N \nu_N^j(U)$, so these probabilities are bounded below by a fixed positive number for large $N$. The uniform upper bound (10.4) on all marks and (10.6) at $\sigma_{j,N}$ would then give

$$J_{\epsilon_N}(z_*) \leq \int J_{\epsilon_N}\, d\nu_N^j \leq -\frac{\alpha}{4\epsilon_N}\nu_N^j(U) + U_B(1 - \nu_N^j(U)) \longrightarrow -\infty.$$

This contradicts (9.4). Thus $W(z_j) = v(z_j)$ almost surely. A countable intersection proves this simultaneously for all $j$. The derivation of (10.17) used unconditional expectations; it does not require conditioning on the flag.

For each fixed $j$ define the subset of $\mathcal{X}$

$$E_j = \{\eta = 1, \quad |x_{\text{end}} - x_j - p_j(t_{\text{end}} - \delta_j)| \le K\delta_j\}\,.$$

This event retains the actual terminal time in the prelimit laws. It is closed: the flag condition is closed and the norm is a continuous function of the terminal point, the $j$-th marked point, and $p_j$. Equation (10.15) gives $\mu_N(E_j) \ge 1/2$ whenever $N \ge j$. The closed-set Portmanteau inequality has the direction

$$\mu(E_j) \ge \limsup_{N\to\infty} \mu_N(E_j) \ge \frac{1}{2}.$$

These events need not be independent. For every $h$ their tail union has measure at least $1/2$, since it contains each $E_j$ with $j \ge h$. Continuity of probability from above therefore gives

$$\mu\left(\limsup_{j\to\infty} E_j\right) = \lim_{h\to\infty} \mu\left(\bigcup_{j\ge h} E_j\right) \ge \frac{1}{2}.$$

Choose one outcome in this limsup event and in the full-measure sets of (10.16)–(10.17). Every $E_j$ includes $\eta = 1$, so this outcome has $t_{\text{end}} = 0$ and $t_j = \delta_j$ for every $j$. It belongs to $E_j$ for an infinite set of indices, at each of which its localization inequality is

$$|x_{\text{end}} - x_j + p_j\delta_j| \le K\delta_j.$$

The same outcome has a finite common slope at the endpoint, contact at all the marks, and the indicated fits. Its endpoint and the infinite set of these marks establish (10.1). □

# 11 Rigidity of the final limit

The propagation result produces contacts at shrinking scales near one terminal point. The finite slope at that point makes the final limit have constant forward and backward envelope slopes at every length. We first show why this forces an affine function.

**Lemma 11.1** (Rigidity from constant envelope slopes)**.** *Let $v \in \mathcal{A}_{m,C_0}$, with the class and envelopes defined in Section 7. Suppose that for some $(b, t_0) \in \mathbb{R}^m \times \mathbb{R}$ and $B \in \mathbb{R}$,*

$$\mathcal{S}_\ell^+ v(b, t_0) = \mathcal{S}_\ell^- v(b, t_0) = B \qquad \text{for every } \ell > 0. \tag{11.1}$$

*Then $v$ is affine.*

*Proof.* Translate $(b, t_0)$ to $(0, 0)$ and subtract the value there; write $V$ for the resulting function. These changes preserve the class and the envelope slopes. The forward equality in (11.1) gives $\mathcal{P}_t V(0, 0) = Bt$ for $t > 0$, and therefore

$$V(y, t) \le Bt + \frac{|y|^2}{2t} \qquad (t > 0). \tag{11.2}$$

Fix any $(x, t) \in \mathbb{R}^m \times \mathbb{R}$. At a large length $\ell$, a maximizing position $y_\ell$ for $\mathcal{P}_\ell V(x, t)$ satisfies

$$|y_\ell| \le C_{x,t}\ell(1 + \log \ell)^2. \tag{11.3}$$

This is the optimizer estimate from Section 7, applied on a box of size comparable to $\ell$; the translated function has the same growth estimates with constants depending on a fixed value

and fit. Explicitly, if the displacement is $b_\ell \ell$ with $b_\ell \geq 2$, comparison with zero displacement and the bound $|V(y,s)| \leq CK(1+\log K)^2$ on boxes of size $K$ give

$$b_\ell \leq C_{x,t}(1+\log \ell + \log b_\ell)^2.$$

Absorbing the logarithmic term in $b_\ell$ gives $b_\ell \leq C'_{x,t}(1+\log \ell)^2$, which proves (11.3).

For $\ell$ sufficiently large, $t+\ell > 0$, so (11.2) applies at the optimizer. The difference between the two quadratic costs is

$$\begin{aligned}\frac{|y_\ell|^2}{2(t+\ell)} - \frac{|y_\ell - x|^2}{2\ell} &= -\frac{t|y_\ell|^2}{2\ell(t+\ell)} + \frac{x\cdot y_\ell}{\ell} - \frac{|x|^2}{2\ell} \\ &= O_{x,t}\big((1+\log \ell)^4\big) = o(\ell)\end{aligned}$$

by (11.3). Consequently

$$\limsup_{\ell\to\infty} \mathcal{S}_\ell^+ V(x,t) \leq B.$$

The forward slopes are nondecreasing in $\ell$ by Lemma 7.4. Thus $\mathcal{S}_\ell^+ V(x,t) \leq B$ for every $\ell > 0$. Evaluating the envelope at any competitor gives the global directed comparison

$$V(y,t+\ell) - V(x,t) \leq B\ell + \frac{|y-x|^2}{2\ell} \qquad (x,y\in\mathbb{R}^m,\ t\in\mathbb{R},\ \ell>0). \tag{11.4}$$

The envelopes are attained by Lemma 7.3. At the origin, (11.1) therefore supplies, for every $\ell, r > 0$, a forward equality point $(\ell q_+, \ell)$ and a backward equality point $(-rq_-, -r)$, with

$$V(\ell q_+, \ell) = B\ell + \tfrac{1}{2}\ell|q_+|^2, \qquad V(-rq_-, -r) = -Br - \tfrac{1}{2}r|q_-|^2.$$

Apply (11.4) directly from the backward point to the forward point. After canceling $B(\ell+r)$, it gives

$$0 \geq \frac{\ell|q_+|^2 + r|q_-|^2}{2} - \frac{|\ell q_+ + rq_-|^2}{2(\ell+r)} = \frac{\ell r}{2(\ell+r)}|q_+ - q_-|^2.$$

Hence every forward and backward equality velocity is one vector $q$, independent of all lengths and choices. We have

$$V(qs,s) = (B + |q|^2/2)s \qquad (s\in\mathbb{R}).$$

Let $A(x,t) = q\cdot x + (B - |q|^2/2)t$. Comparing $(x,t)$ to a point $(qs,s)$ on this line in (11.4) yields

$$V(x,t) \leq A(x,t) + \frac{|x-qt|^2}{2(t-s)} \quad (s<t), \qquad V(x,t) \geq A(x,t) - \frac{|x-qt|^2}{2(s-t)} \quad (s>t).$$

Letting $s\to-\infty$ and $s\to+\infty$ proves $V = A$. □

*Proof of Theorem 1.3.* Suppose that the class contains a nonaffine function. Proposition 8.4 supplies a centered solution $v$, a number $\kappa > 0$, and its subsolution envelope

$$W(x,t) = \mathcal{C}_t v(x,t) - \kappa^2 t/2 \qquad (t>0)$$

with the stated contact and finite-slope properties. Proposition 10.1 supplies a point $x_{\mathrm{end}}$, infinitely many indices $j$ with $r_j = 2^{-j}$, contacts $(x_j, r_j)$, and fits $(p_j, a_j)$ there of radius $r_j$, such that

$$H_v(x_{\mathrm{end}}, 0) > -\infty, \qquad W(x_j, r_j) = v(x_j, r_j), \qquad |x_{\mathrm{end}} - x_j + p_j r_j| \leq K r_j \tag{11.5}$$

for one finite $K$.

Apply Lemma 9.3 at each of these marks, and denote the transformed functions by $\bar{v}_j$. Their images of $(x_{\mathrm{end}}, 0)$ are $(b_j, 0)$, where

$$b_j = \frac{x_{\mathrm{end}} - x_j + p_j r_j}{r_j}.$$

These vectors are bounded by (11.5). Every $\bar{v}_j$ has value zero and the zero fit of radius one at $(0, 1)$. Lemma 7.2 gives a subsequence for which $\bar{v}_j \to v_\infty$ locally uniformly and $b_j \to b$. The uniform growth bounds also give convergence of the forward and backward envelopes at each fixed positive length, by Lemma 7.3.

The slope covariance (9.14) gives

$$\mathcal{S}^{\pm}_{\ell} \bar{v}_j(b_j, 0) = \mathcal{S}^{\pm}_{r_j \ell} v(x_{\mathrm{end}}, 0) - a_j \qquad (\ell > 0).$$

Let $B = \mathcal{S}^{+}_{1} v_\infty(b, 0)$, which is finite. For each fixed $\ell > 0$, subtract the equality with the forward slope at length one:

$$\begin{aligned} \mathcal{S}^{\pm}_{\ell} \bar{v}_j(b_j, 0) - \mathcal{S}^{+}_{1} \bar{v}_j(b_j, 0) &= \mathcal{S}^{\pm}_{r_j \ell} v(x_{\mathrm{end}}, 0) - \mathcal{S}^{+}_{r_j} v(x_{\mathrm{end}}, 0) \\ &\longrightarrow 0. \end{aligned}$$

The limit is zero because both unshifted slopes tend to the same finite value $H_v(x_{\mathrm{end}}, 0)$. Passing to the limit on the left gives

$$\mathcal{S}^{+}_{\ell} v_\infty(b, 0) = \mathcal{S}^{-}_{\ell} v_\infty(b, 0) = B \qquad (\ell > 0).$$

This argument uses each fixed length along the same locally convergent subsequence; no convergence of the individual shifts $a_j$ is required. Lemma 11.1 now makes $v_\infty$ affine.

The contacts retain a positive envelope gap. By (8.9) at the contacts in (11.5) and the gap covariance (9.13),

$$\mathcal{C}_1 \bar{v}_j(0, 1) - \bar{v}_j(0, 1) = \kappa^2/2.$$

Lemma 8.1 passes this identity to the locally uniform limit with its uniform growth bound. Thus

$$\mathcal{C}_1 v_\infty(0, 1) - v_\infty(0, 1) = \kappa^2/2 > 0.$$

For an affine function, the average of its values at any configuration is its value at the barycenter, so the nonnegative variance penalty makes $\mathcal{C}_1 v_\infty = v_\infty$. This contradiction proves Theorem 1.3. The reduction in Section 5 then proves Theorem 1.2. □

# 12 Conclusion

For each $d \geq 2$, the two viscosity inequalities $-M \leq \Delta_\infty u \leq M$ imply an interior $C^{1,\alpha_d}$ estimate with constants depending only on $d$ and with the scale-invariant size $\mathrm{osc}_{B_R} u/R + (RM)^{1/3}$. The exponent is positive and can be chosen at most $1/3$. The normalized operator satisfies the corresponding estimate with size $\mathrm{osc}_{B_R} u/R + RM$.

The forcing enters the proof through two explicit quantities: a bounded error in the three-point jet inequality, and $r_i/(\lambda_i c_i^3)$ in the rescaled equation. The first is dominated by the negative curvature of the comparison penalty, while the selected scales force the second to vanish. The resulting entire limit satisfies the sheared affine estimates of Theorem 1.3. Its rigidity closes the power estimate, and the fitting slopes converge with a uniform Taylor remainder. This proves the stated regularity without any continuity assumption on a separately specified forcing term.